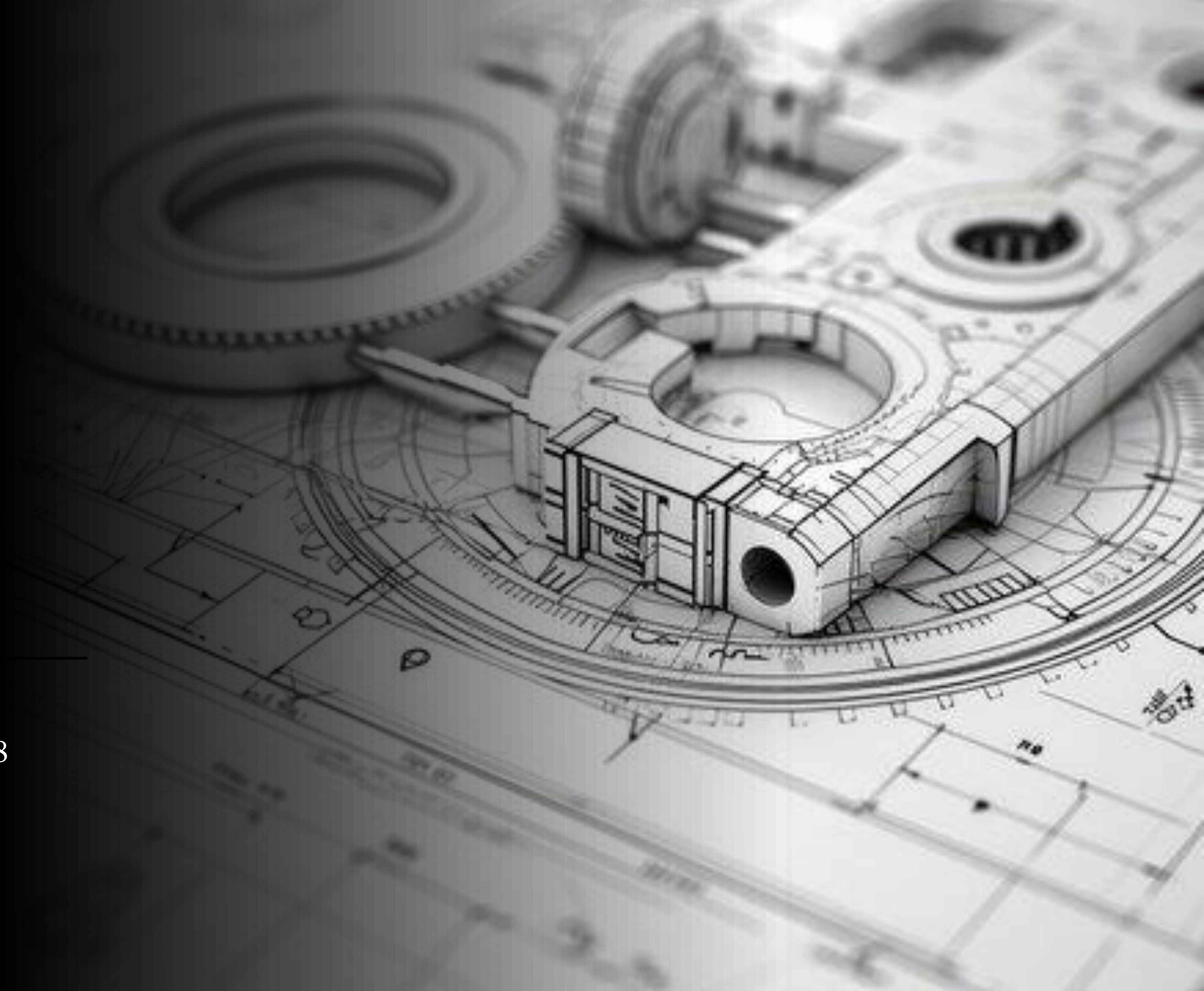
Yuvrajsinh Thakor
M.Eng (Mechanical)
B.E (Mechanical)

Mechanical Design Portfolio



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Welcome, and thank you for taking the time to view my portfolio. The Goal of this portfolio is to give you a deeper insight into my experiences and skills I have gained over my recent history.

It is my hope that this will allow you to better assess how my skills can be applied to your company.

I would be happy to talk in more detail and can be reached using the contact information on the first page.

About me

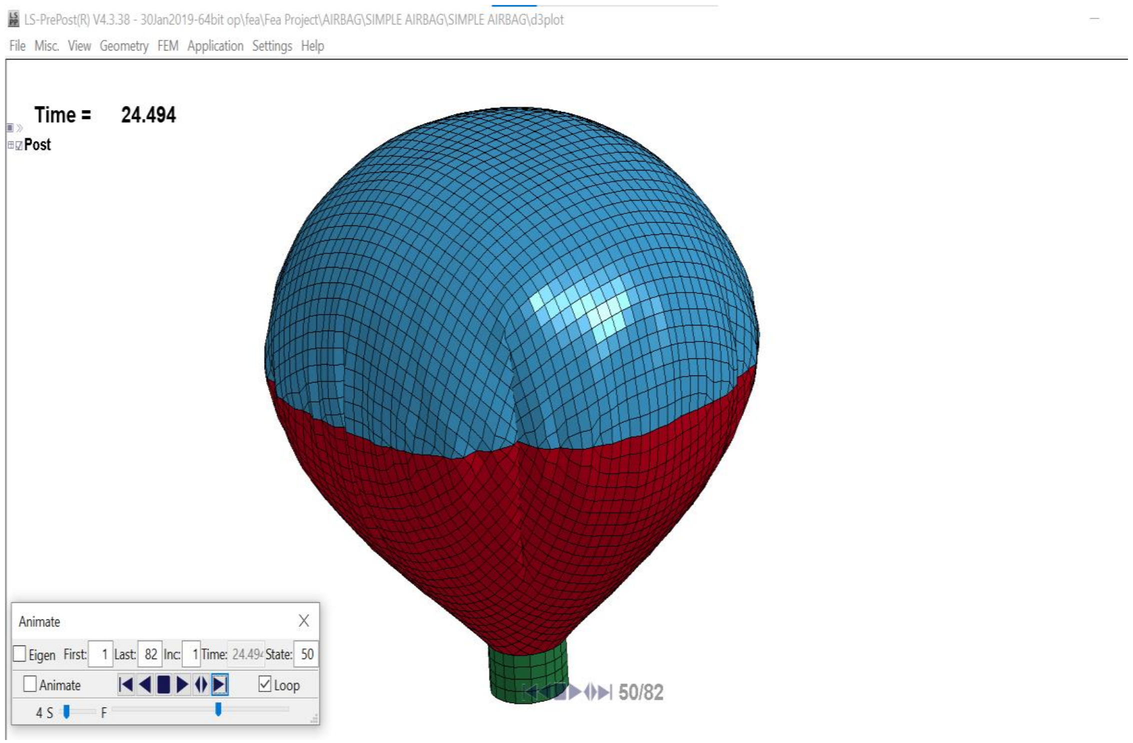
- From a young age, I’ve always been captivated by machines and the art of turning ideas into functional designs. My journey as a Mechanical Engineer officially began in 2015 when I pursued my Bachelor's in Mechanical Engineering. This foundational step fueled my passion for both the intricate details of machine mechanics and the creative process behind designing them.
- Throughout my career, I’ve had the opportunity to apply my skills in real-world settings. At **Apeshwar Engineering Works**, I contributed to the **design and optimization of machinery, including paver block machines and plastic pulverizers**. This hands-on experience sharpened my technical expertise, as I worked closely with multidisciplinary teams to tackle complex design challenges and ensure smooth production processes. I honed my proficiency in **SolidWorks, CATIA, and AutoCAD**, tools that became my second language in bringing designs to life.
- As I continued to grow in my field, I realized that to truly push the boundaries of mechanical design, I needed to deepen my knowledge and technical capabilities. That’s why I pursued my **Master’s in Mechanical Engineering** at the University of Windsor, where I further refined my skills in **design simulation, advanced manufacturing processes, and finite element analysis**. This academic journey empowered me to take on even more complex challenges, including designing and simulating airbag systems and advanced steam engines.
- Now, I’m eager to take my skills to the next level by contributing to innovative and sustainable engineering solutions in the design field. My background has equipped me with the ability to tackle complex design challenges, streamline processes, and collaborate effectively with teams to create solutions that make a lasting impact.

Project Title: Design and Analysis of Airbag Deployments and Responses

Aim: To design model and analyze by varying different Variables for a particular model for airbags folding taken from a real model so that the airbags will not fail for a particular time – state response.

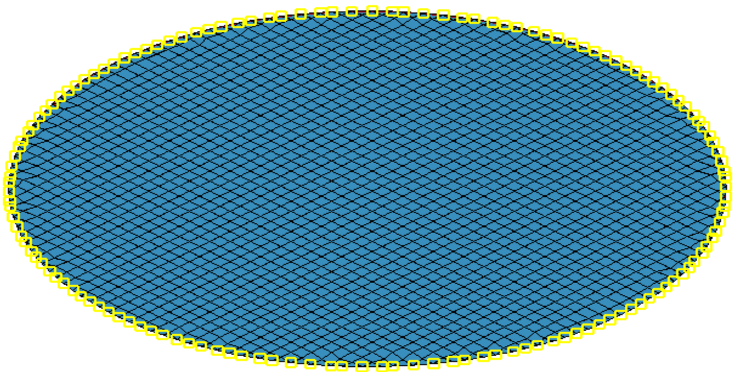
Approach:

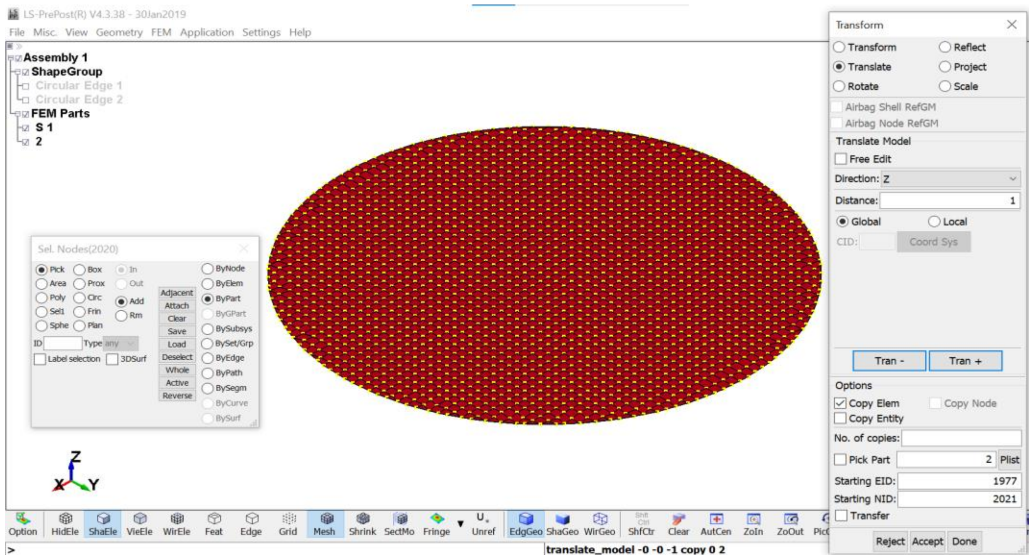
- First all the material properties and parameters of airbag and airbag cover, etc. are calculated and defined.
- In the designing we also calculated all the pressure and loads on airbag and eventually on the complete Airbag which included folding and unfolding of airbags.
- Designing also included selection of the material for the airbags.
- The modeling in the project was carried on LS- PREPOST V4.3 and analysis on LS- DYNA RUN.
- The pressure acting on the airbag is also varied and graphs are plotted to see the pressure and time state response behavior.
- The plotted graph can be used to verify the airbag folding for different favorable conditions. This was developed as a table which shows all the favorable conditions for different time – state response.
- Unfolding process of airbag were also determined.

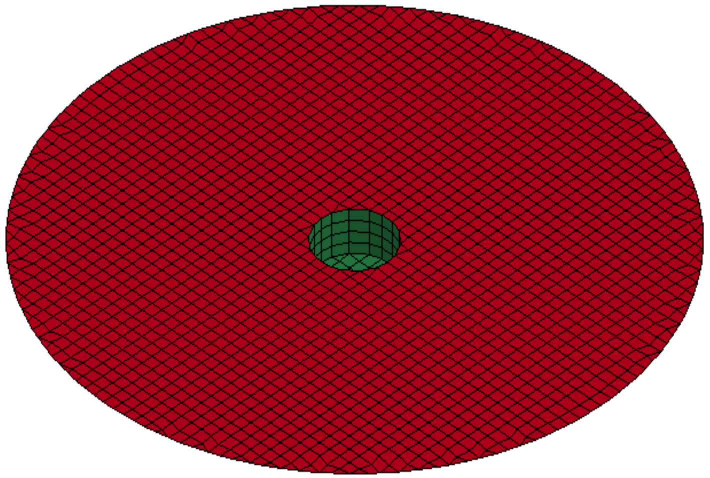


Steps to prepare airbag:

- First the circle of 600 mm and 40 mm is drawn by defining circle function.
- Using of Mesh function, blank masher option mesh is created.
- Translation of finite element part 1 in z direction is done.
- Edge is defined by translating the edge at 45˚ .
- Airbag is formed by merging duplicate nodes, small circle is trimmed and then element is created by using small trimmed translated circle.



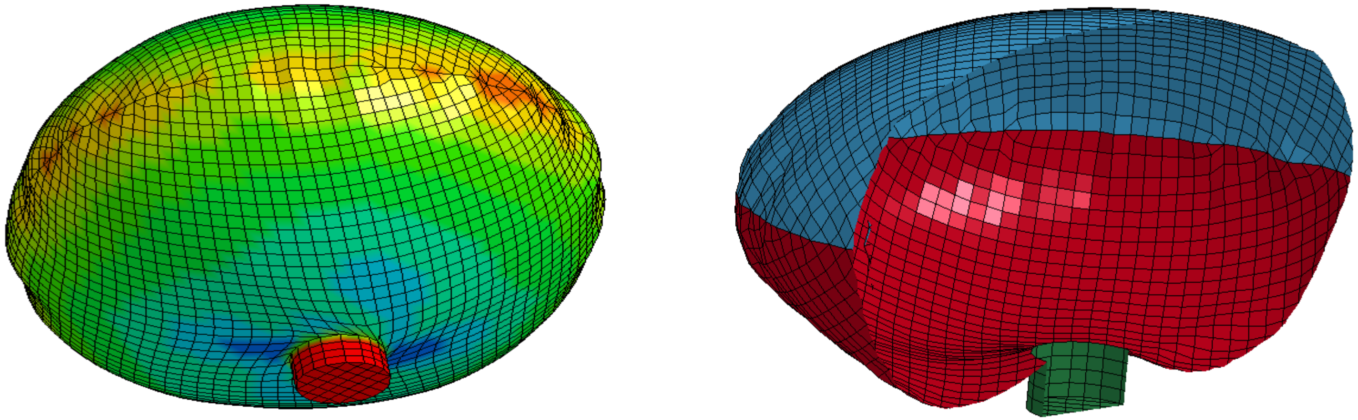




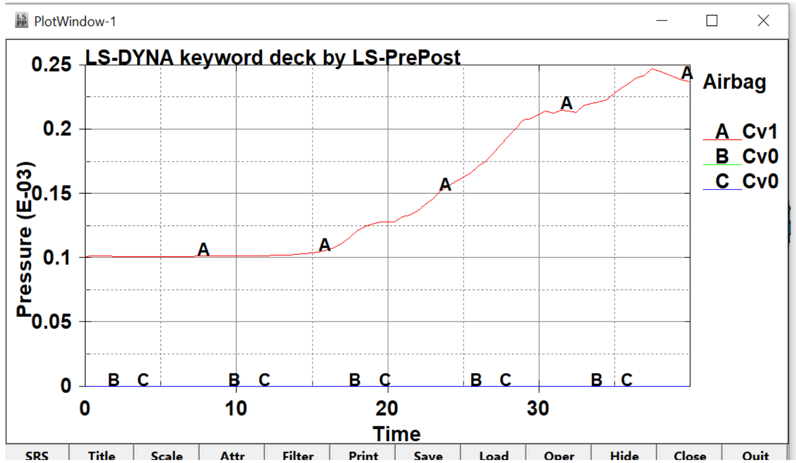
- 034_FABRIC material is applied by defining material properties such as Young’s Modulus E, Mass Density (ρ) , Poisson’s Ratio (ν), Shear Modulus (GAB), Compressive Stress Elimination, Young’s Modulus for Elastic Linear, Poisson’s Ratio for Elastic Linear, Linear to Total Fabric Thickness Ratio and Rayleigh Damping Coefficient.
- 020_RIGID material is applied for inert gas by defining parameters Young’s Modulus, Mass Density, Poisson’s Ratio and Center of Mass Constraint.
- Once the material and section is defined, Part is generated.

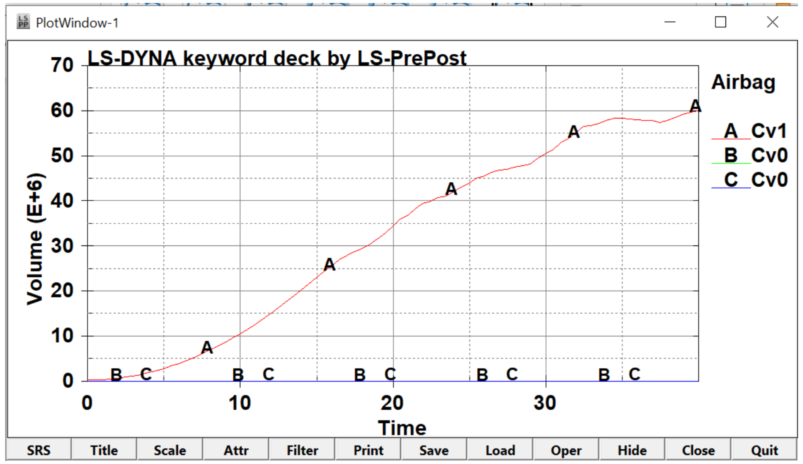
Analysis:

- By taking a particular material for the airbag and by varying its module, the different load were determined for the airbags. The boundary condition is defined by setting free edge, simple airbag model, etc. Preprocessing functions is defined by describing Termination control function to measure time – state response. ASCII function is defined for airbag statistics for plotting graph.
- This value again done for various materials and all the different values were determined.
- These values were them plotted into graphs and the results were analyzed.
- A complete graph was plotted by which airbag with its preferable conditions could be selected by analyzing all the variables.



Isometric and section view of Airbag Unfolding





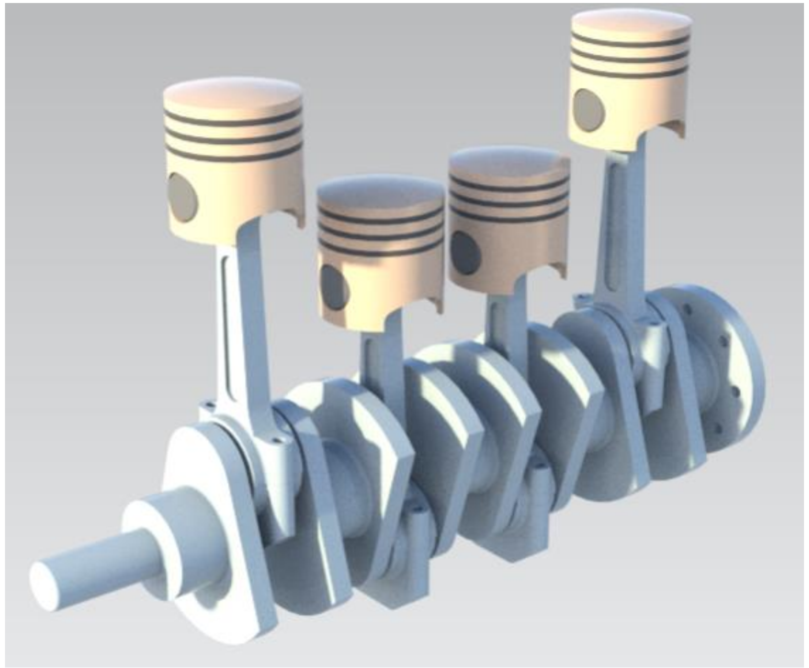
The graph shows different pressure and volume acting on the airbag for give time - state response

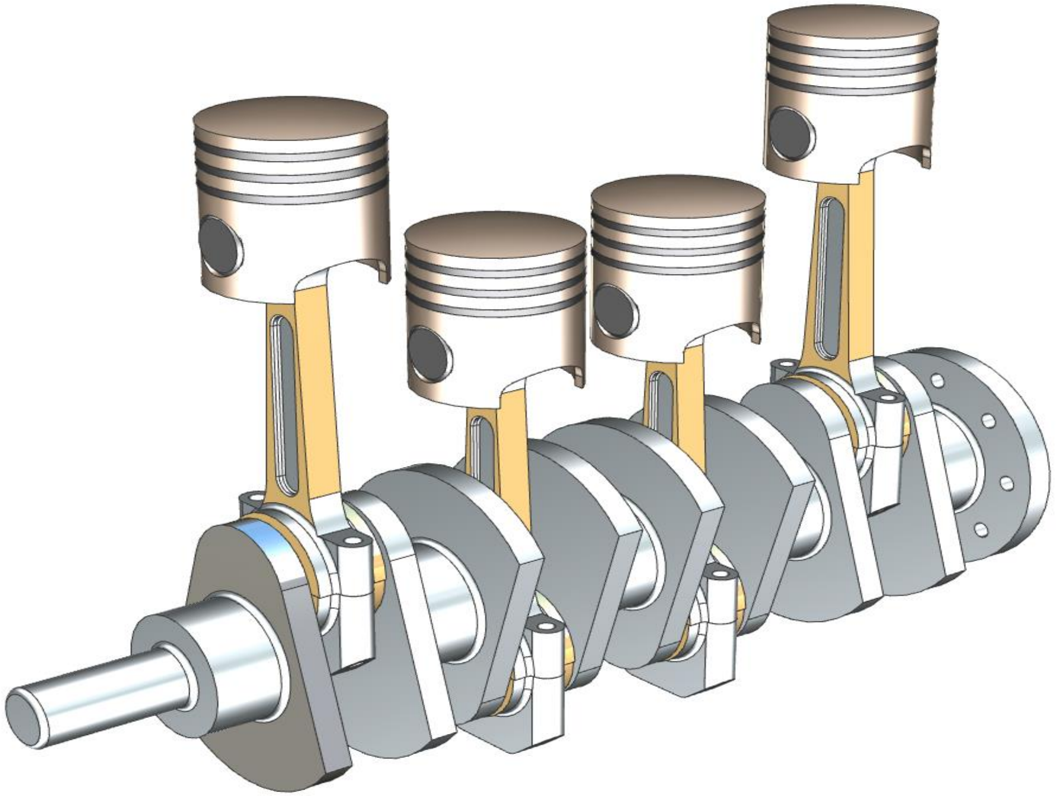
Results:

The complete design with all specifications of the airbag were determined. A complete working model was developed in LS PREPOST with all its specifications. Analysis was successfully completed with all its results collected and presented in the form of charts. The pressure were also calculated with the help of LS DYNA. The pressure with respect to the module, material and volume can be seen from the graphs and a complete chart was created by airbag with its preferable conditions should be selected by analyzing all the variables affecting it.

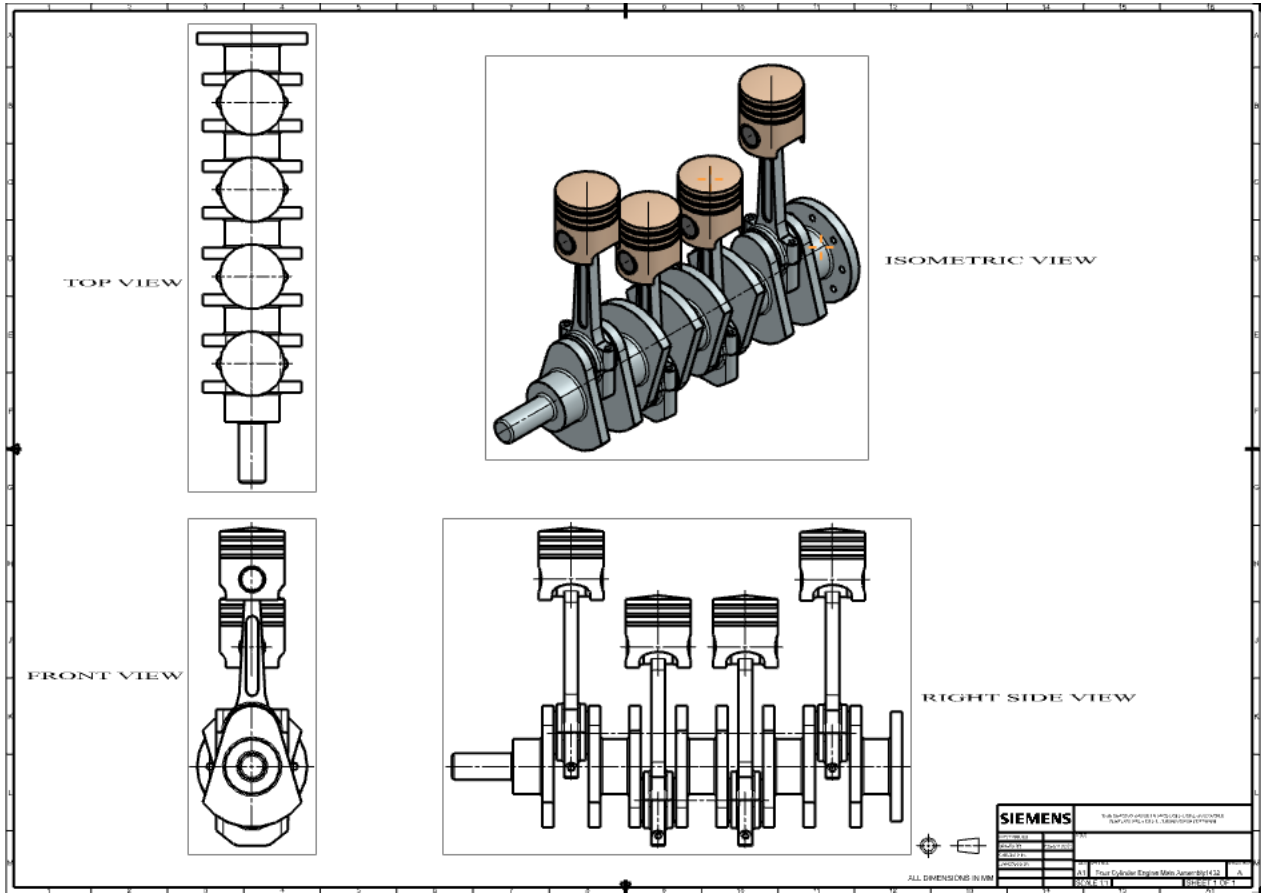
Project Title: Design and Assembly of Four-Cylinder Engine

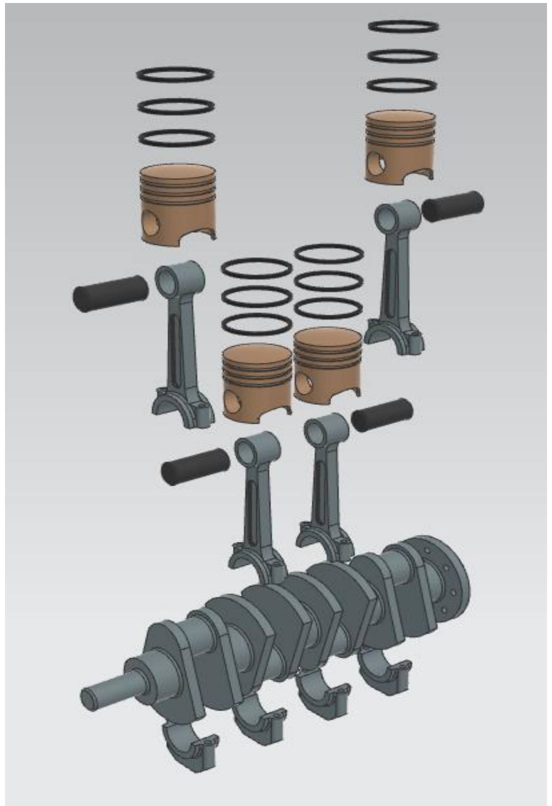
Aim: To design the four-cylinder Engine with all its components with assembled final product and with including Drawings.

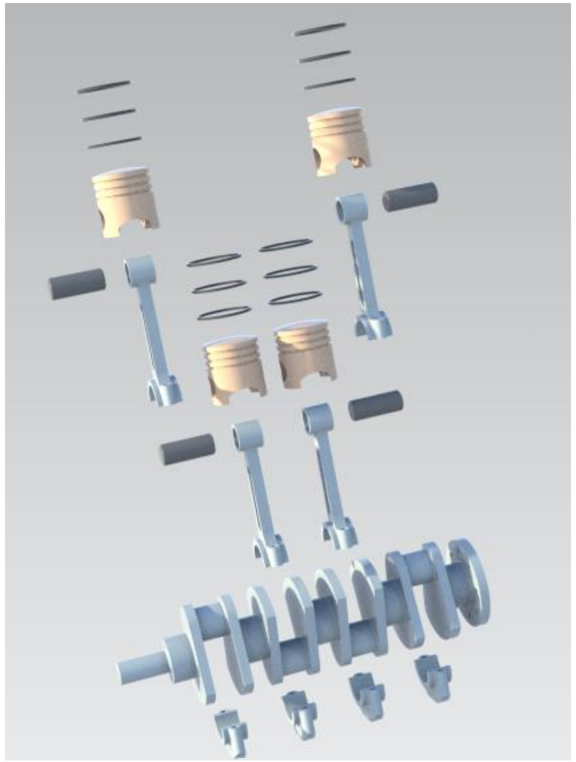




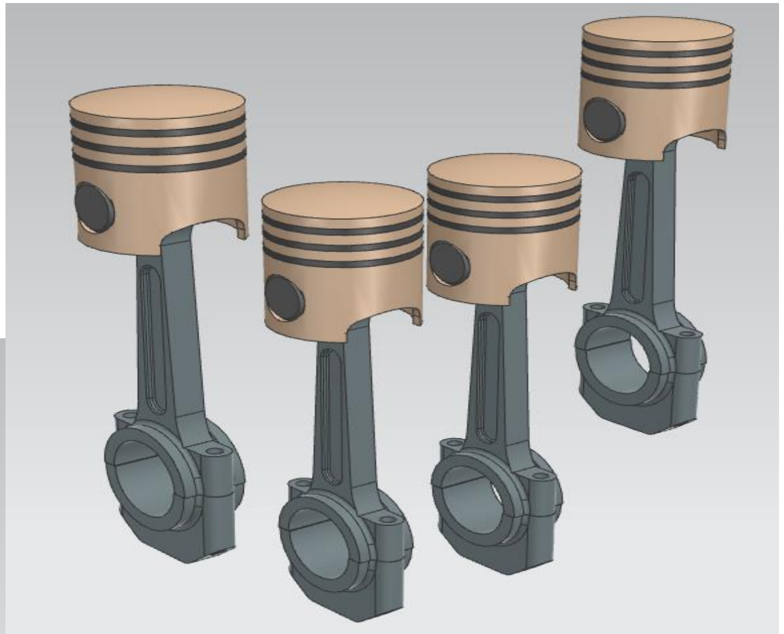
The figure shows the drafting of the main assembly and render view of the exploded view, which shows how the all components assembled together with using of different assembly constraints.



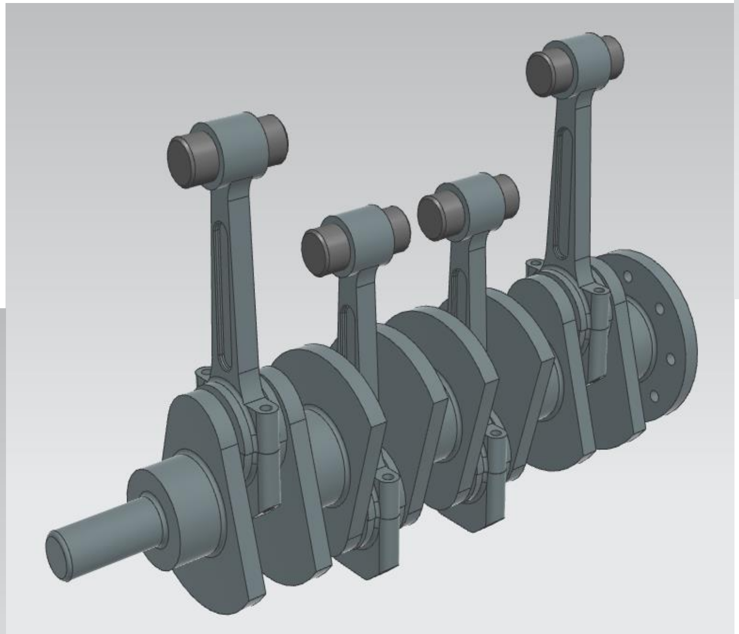




This shows the render version of the assembly without the some of the components.

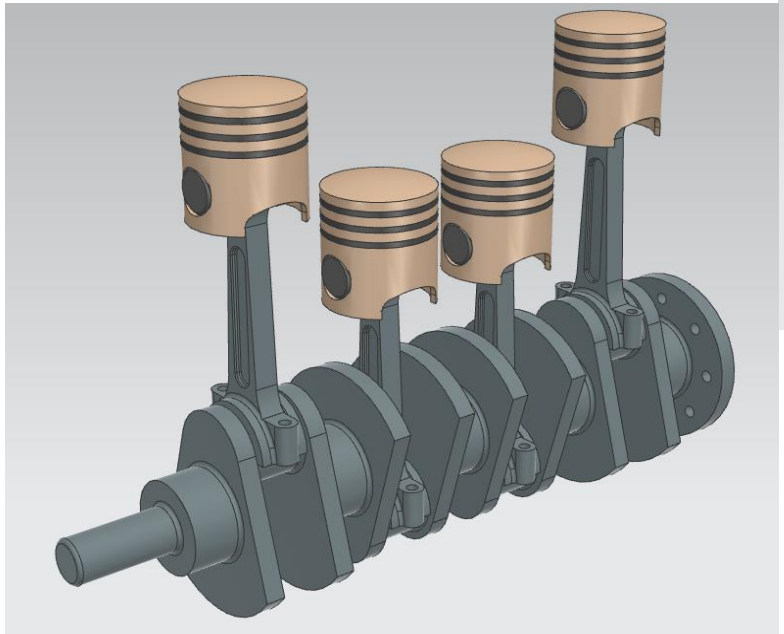


The render version of image without crankshaft.

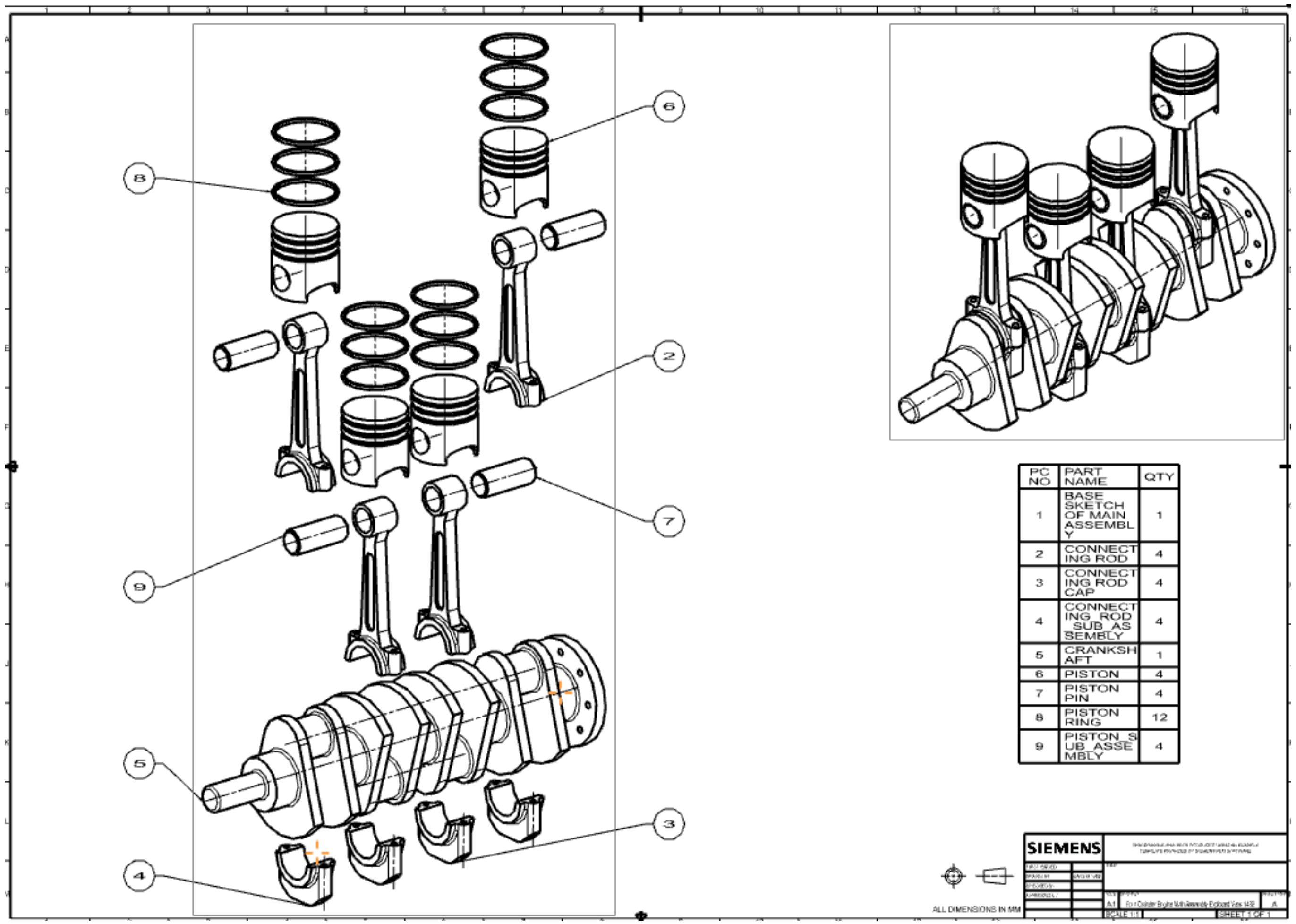


The render version of image without piston and piston ring.

The render version of image without connecting rod cap.

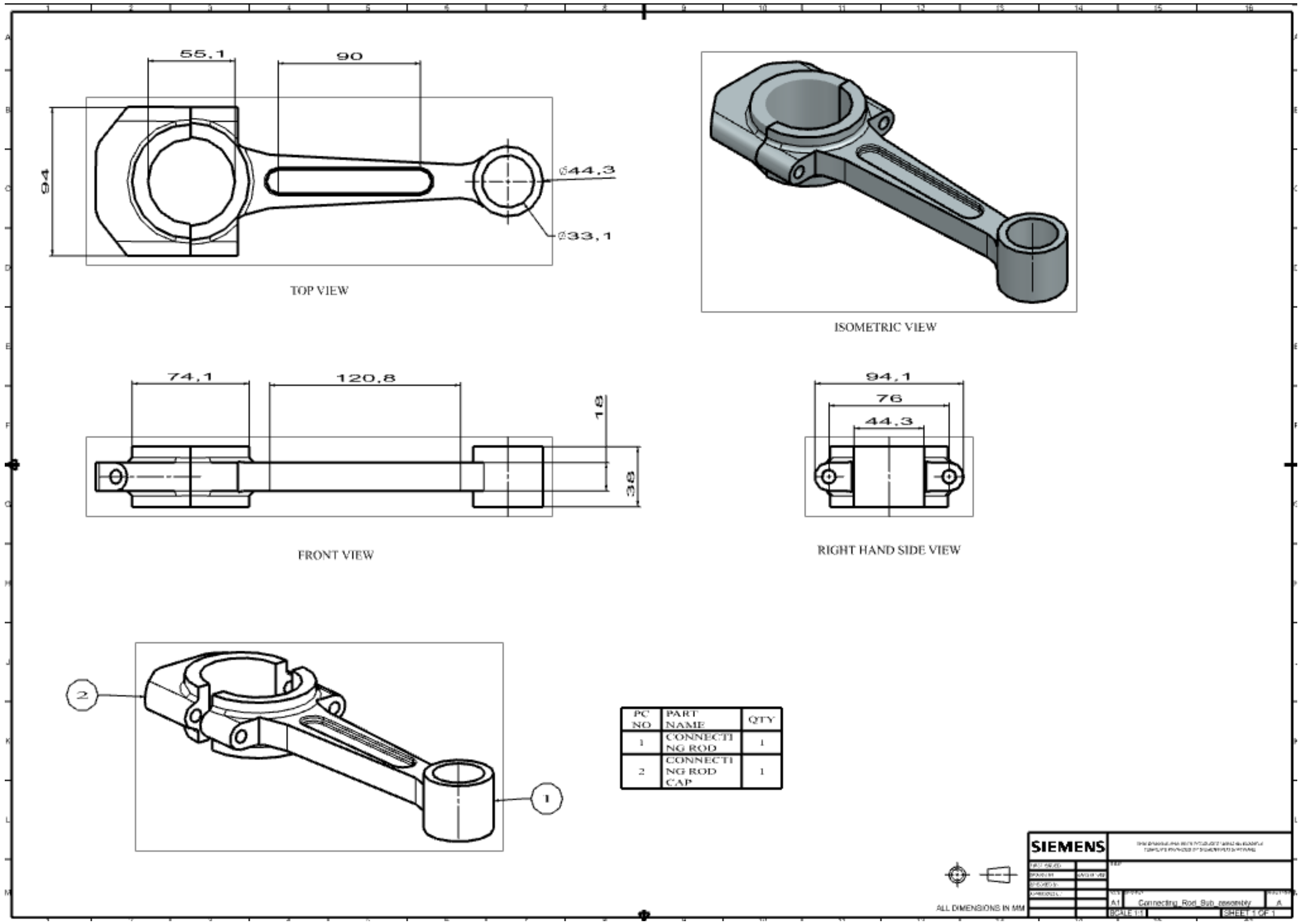


Drafting of exploded view with the BOM (Bill Of Materials). This is detailed view of the component explosion with part list and operation.

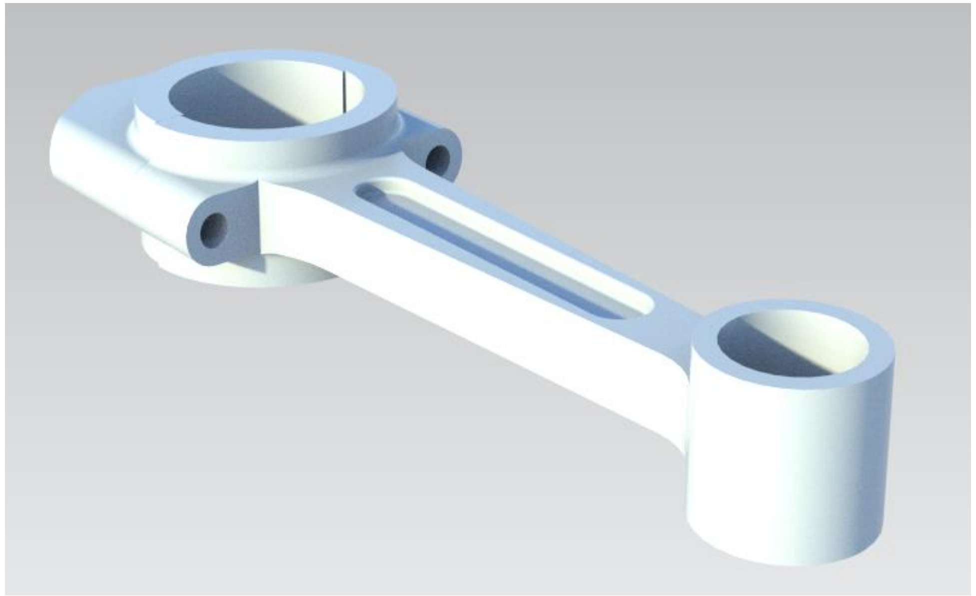


PC NO	PART NAME	QTY
1	BASE SKETCH OF MAIN ASSEMBLY	1
2	CONNECTING ROD	4
3	CONNECTING ROD CAP	4
4	CONNECTING_ROD _SUB_ASSEMBLY	4
5	CRANKSHAFT	1
6	PISTON	4
7	PISTON PIN	4
8	PISTON RING	12
9	PISTON_SUB_ASSEMBLY	4

Connecting Rod Sub Assembly Drafting of different views and BOM which used to connect the crankshaft and piston sub assembly.

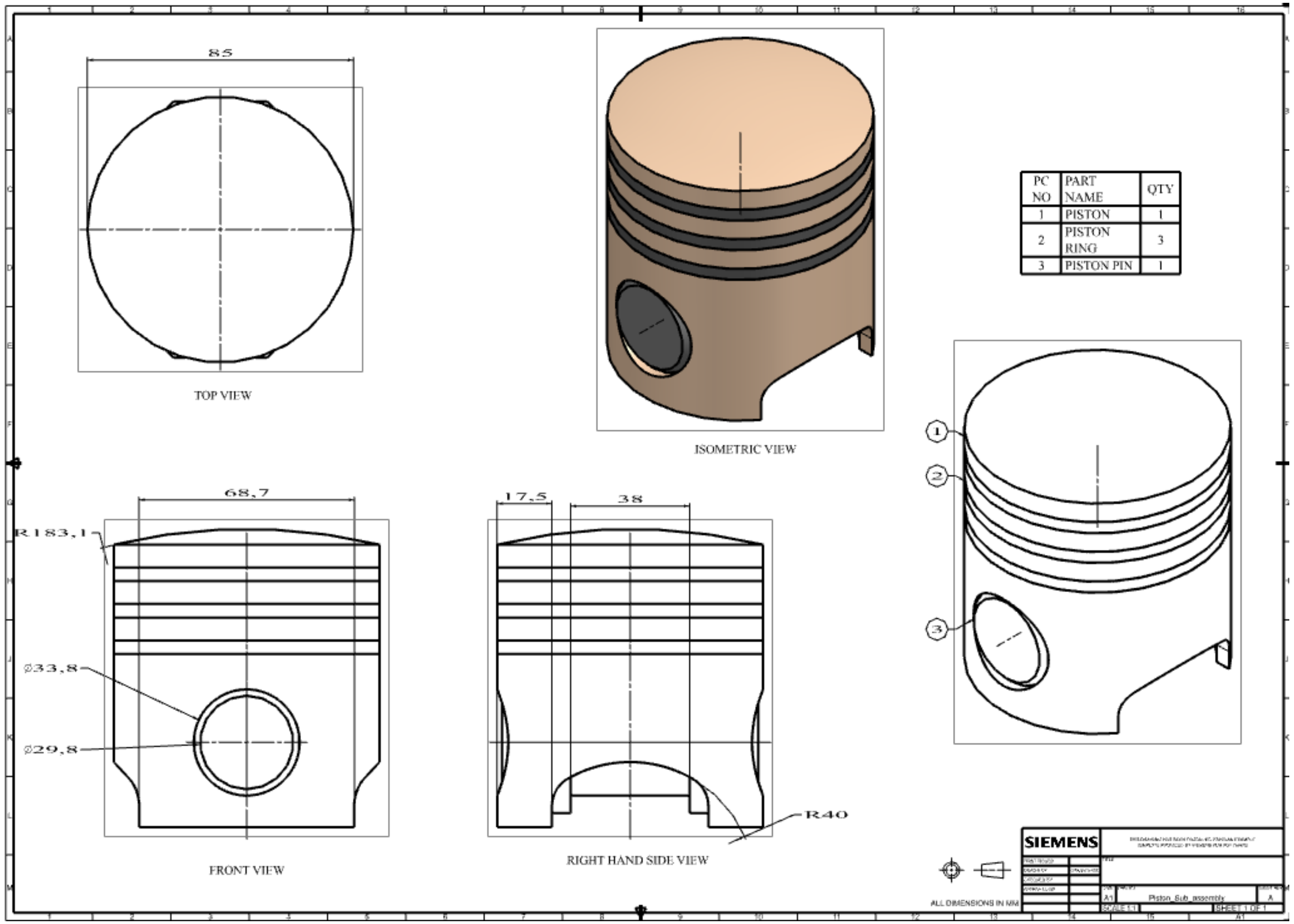


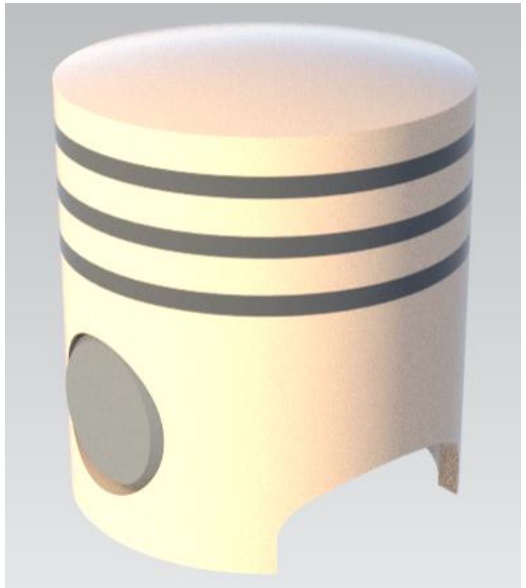
PC NO	PART NAME	QTY
1	CONNECTING ROD	1
2	CONNECTING ROD CAP	1



Connecting Rod Sub Assembly Render View Image version.

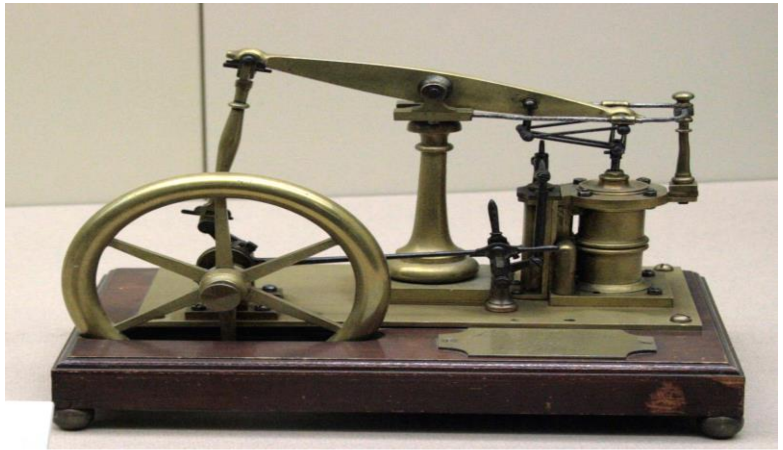
Piston Sub Assembly Drafting of different views and BOM which used to connect the piston and ring and piston pin.





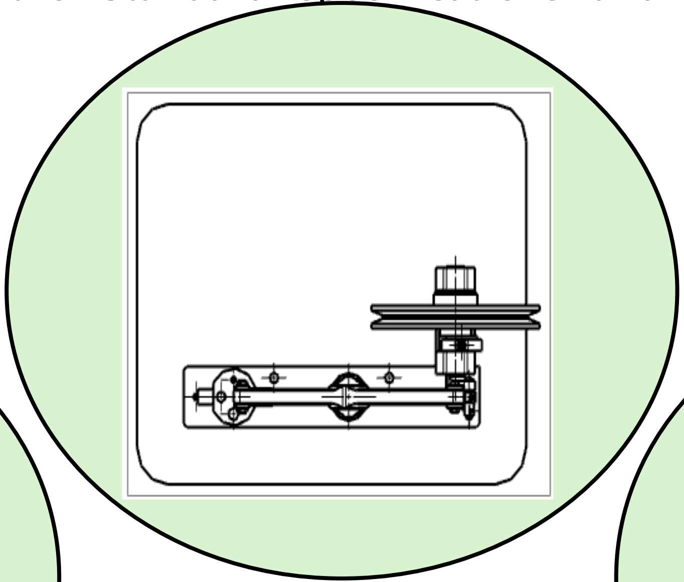
Piston Sub Assembly Render View Image version.

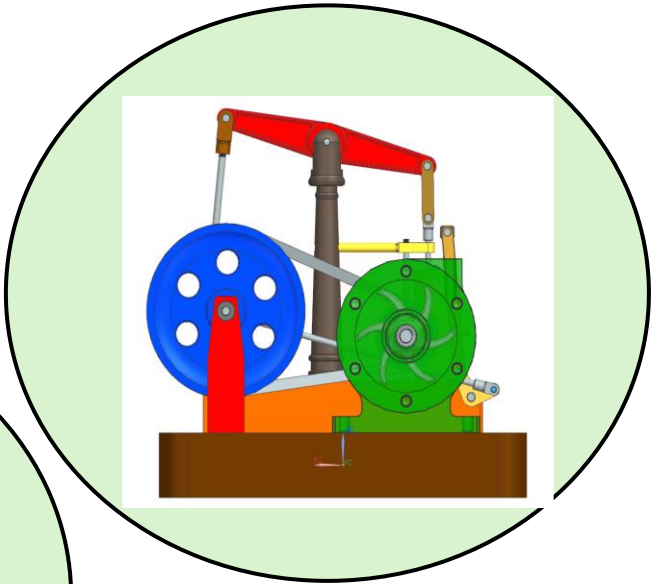
Project Title: Design and assembly of Steam Engine with Doubled Dimensions

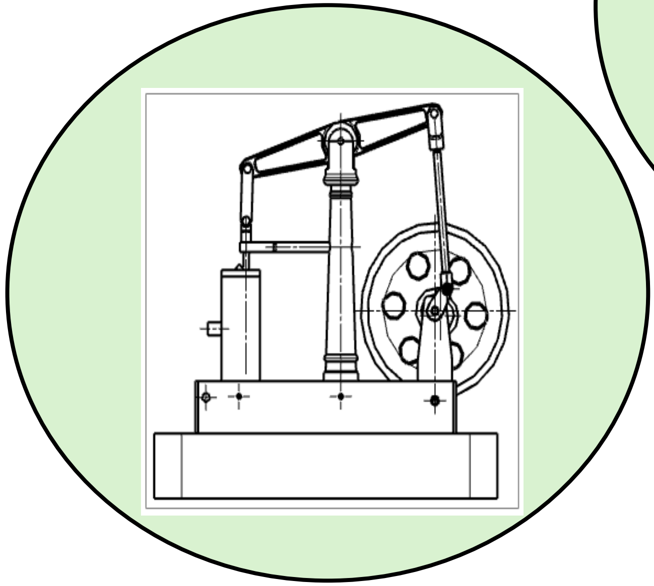


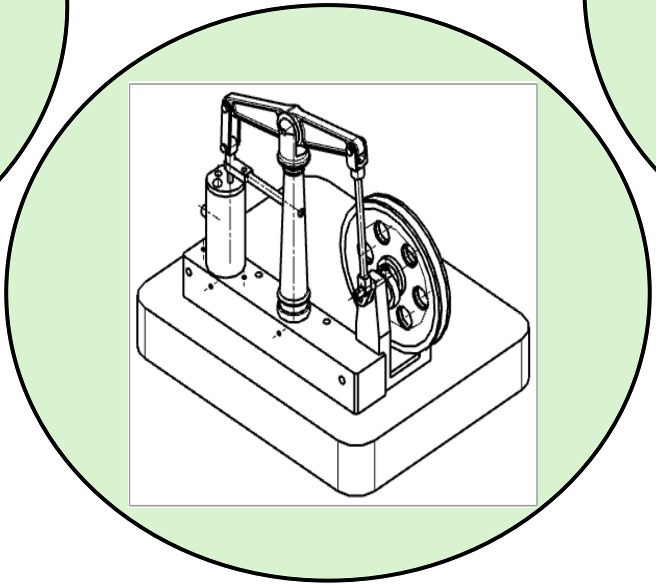
Aim: To design and build the steam engine for our team for final graduation project.

My responsibilities: To design the framework and other components of the steam engine taking into account the Standard specifications and rules of design guidelines.





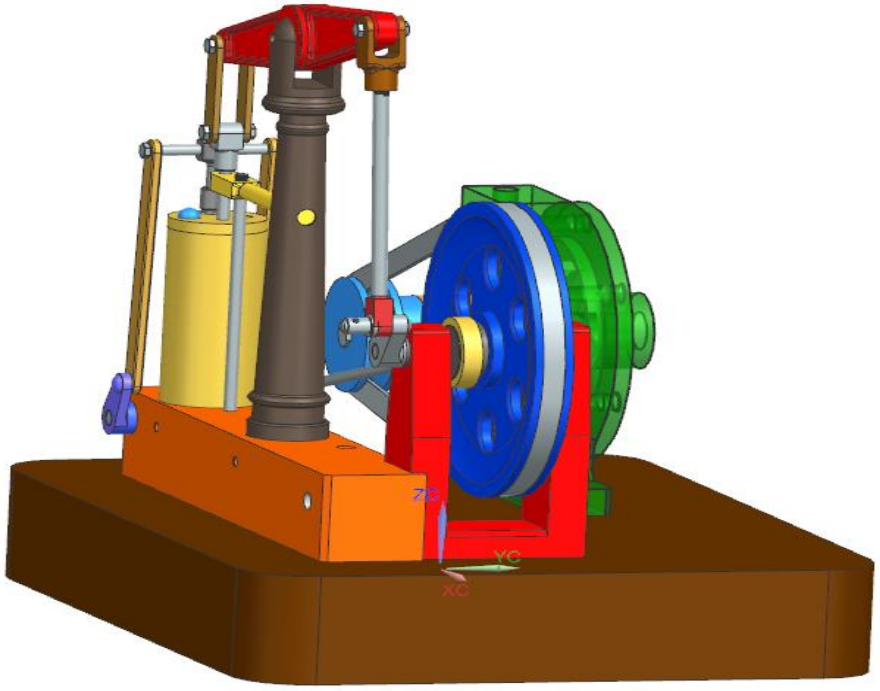




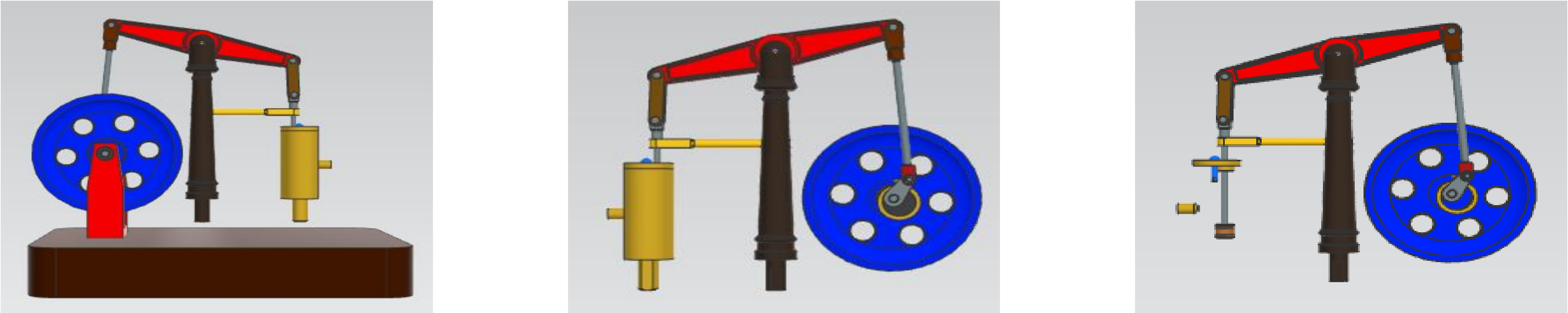
Approach:

First of all, the components such as engine base, engine block, beam linkage, beam, bearing support, centrifugal pump bearing and pulley, centrifugal pump housing, centrifugal pump impeller, cylinder, eccentric hub, piston, plain bearing, turbine pulley, valve linkage dual rocker, valve, sub assembly of crank-shaft, etc. are created using various Boolean operations such as extrude, hole, unite, subtract, revolve, edge blend, chamfer, pattern, mirror features.

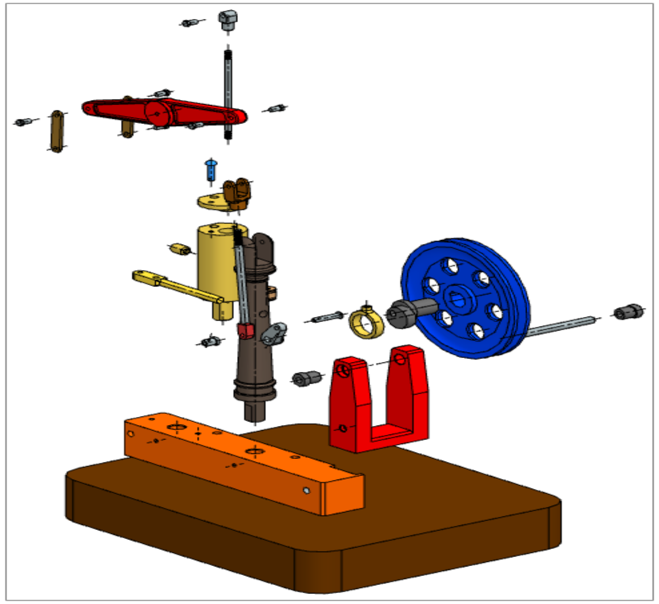
The fig. shows the final assembly of steam engine by assembled various components. The components are assembled by using various assembly constraints such as concentric, touch align, parallel, fixed, etc. which allows the assembly to work freely.



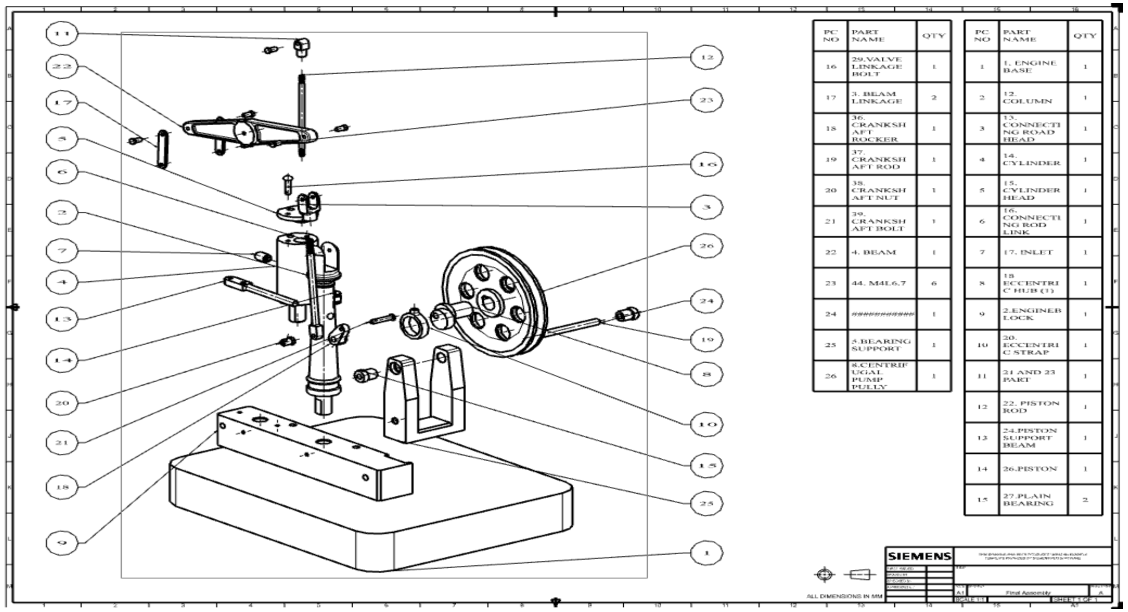
Results: A complete doubled dimensioned steam engine was designed including the assembled product and Drawings and Presentable Rendering with motion analysis.



The above images are the rendered version, with and without the engine base, cylinder, bearing support and eccentric hub.

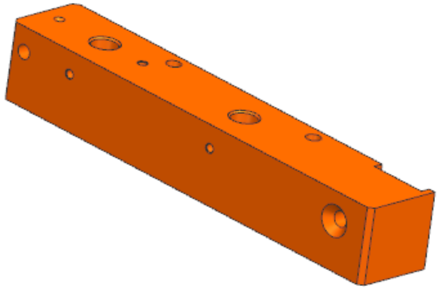


The above image shows exploded view of the steam engine.

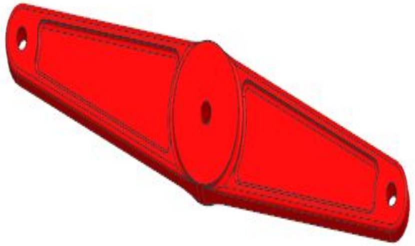


The above image is the exploded view shown as a Drawing in NX Simens with BOM included.

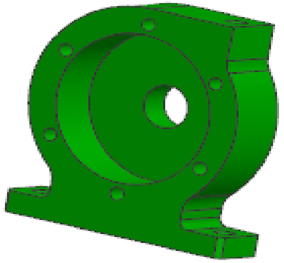
Steam Engine Components:



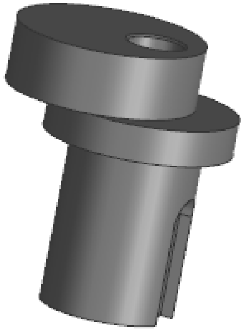
Engine Block



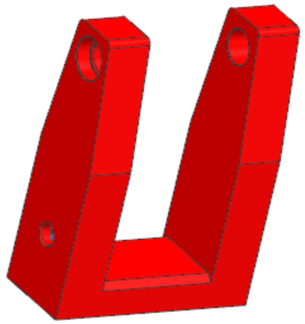
Beam



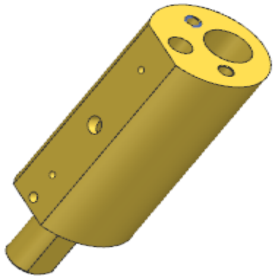
Centrifugal Pump Housing



Eccentric Hub



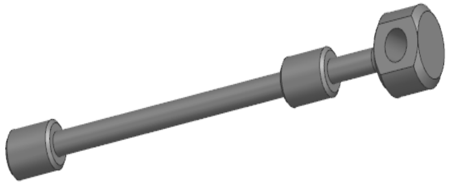
Bearing Support



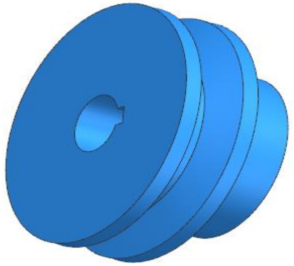
Cylinder

Centrifugal Cover

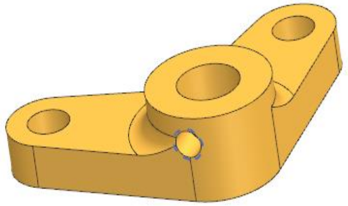
Centrifugal Pump Impeller



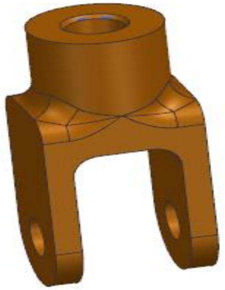
Valve



Turbine Pulley

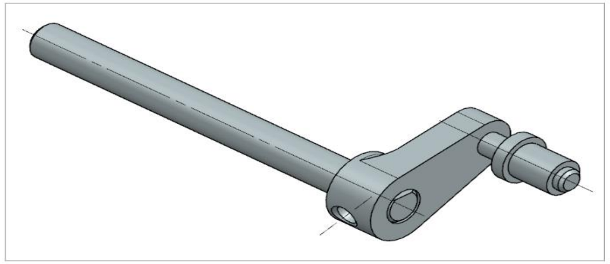


Valve Linkage Dual Rocker



Cylinder Head

Centrifugal Pump Pulley



Crankshaft

- **Project Title: Design of Pressure Vessel and FEA Analysis**
- **Project Objectives:**
  - Designed a boiler pressure vessel adhering to ASME Section VIII standards, ensuring structural safety and compliance with code requirements.
  - Conducted finite element analysis (FEA) to evaluate stress, fatigue, and deformation under operational conditions.
- **Approach:**
  - **Design Calculations and Material Selection:** Calculated shell, head, and nozzle thicknesses using ASME Section VIII formulas. Chose Stainless Steel 316 for its excellent corrosion resistance, weldability, and ability to withstand high pressure and temperature.
  - **SolidWorks Modelling:** Created a 3D model of the pressure vessel, including ellipsoidal heads and reinforced manholes, ensuring design precision. Developed a manhole and nozzle system to enhance functionality while maintaining structural integrity.

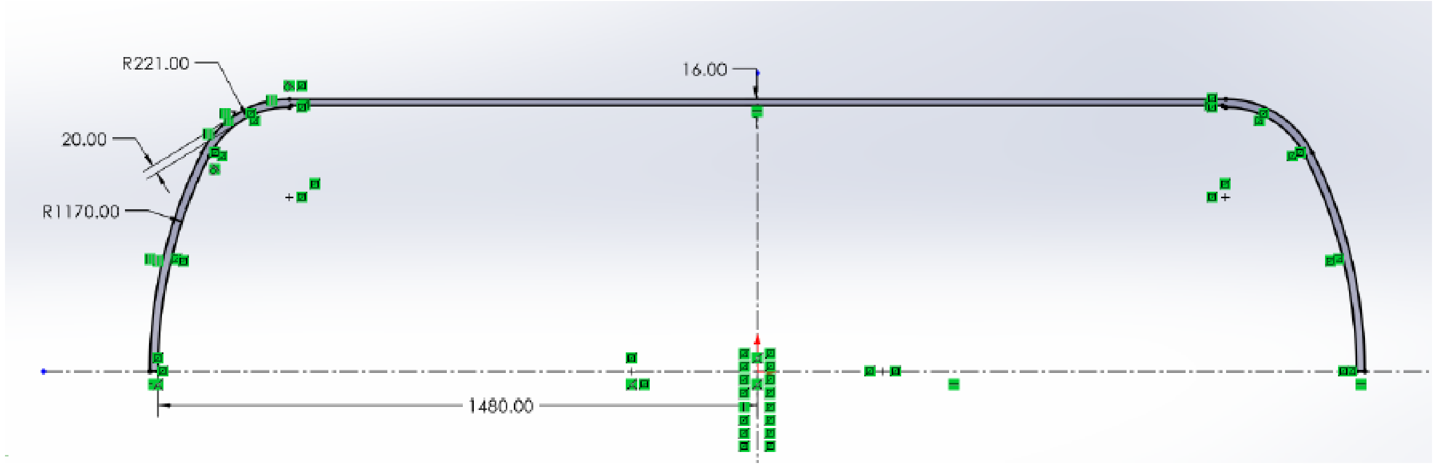


Figure 1: Shell dimensions in SolidWorks

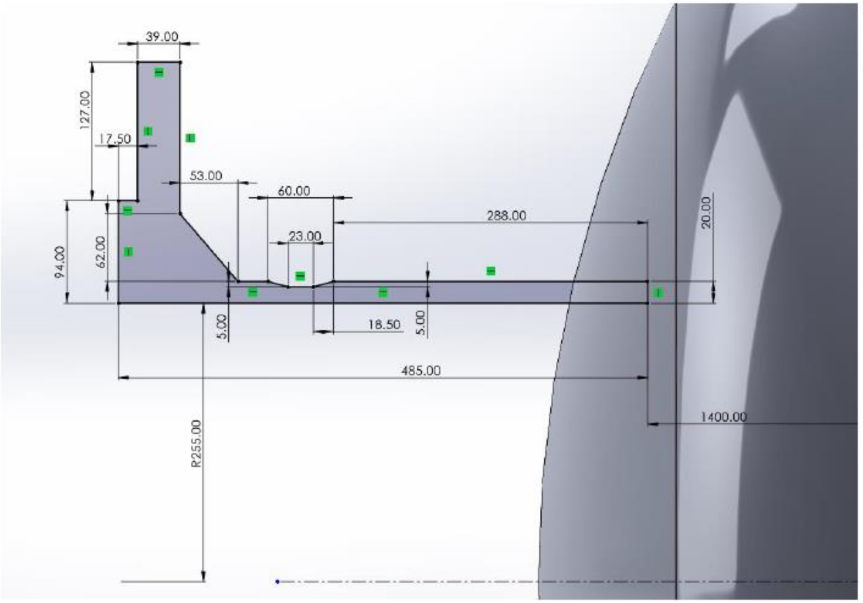


Figure 2: Manhole dimensions in SolidWorks

- **Approach:**
  - **Finite Element Analysis:** Conducted static, fatigue, and hydrostatic analyses in SolidWorks Simulation to assess stress concentrations near openings and junctions. Fatigue life and safety factors under cyclic loads. Verified the vessel's maximum allowable working pressure (MAWP) of 3.6 MPa.

4.2 Static Analysis (Pressure):

Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	6.293e+04N/m^2 Node: 4885	1.127e+09N/m^2 Node: 6529

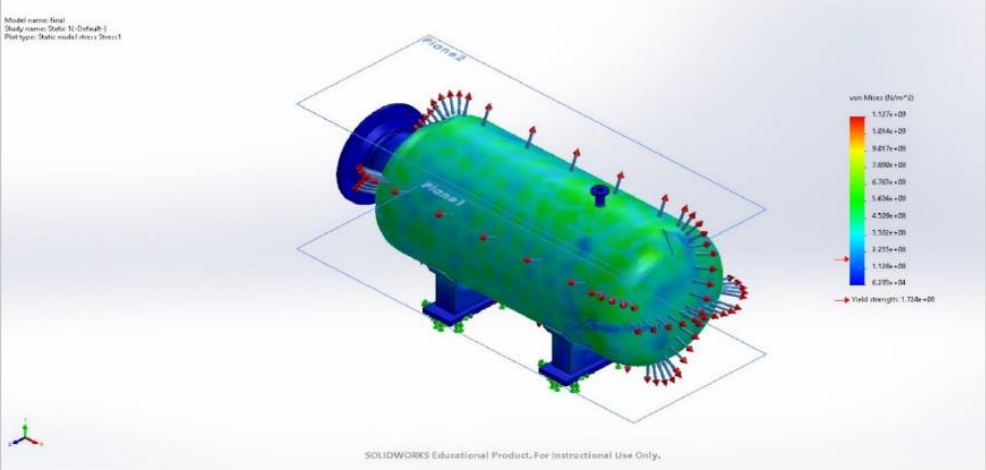


Figure 4: Static 1-Stress-Stress1

4.1 Static Analysis

Model Information

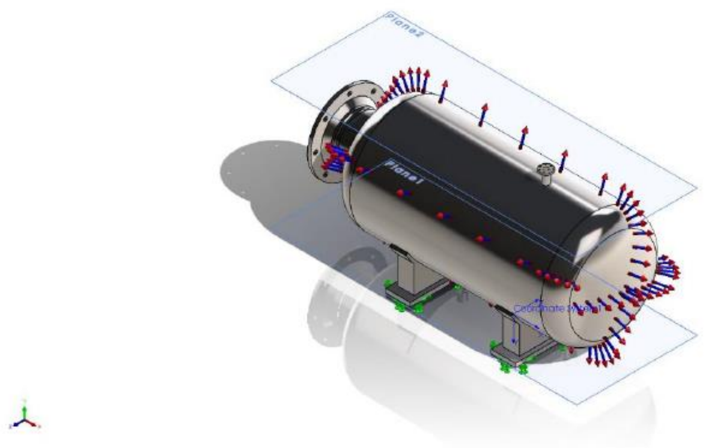


Figure 3: Model name: Pressure Vessel
Current Configuration: Default

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	3.770e-07 Element: 1252	3.111e-03 Element: 6153

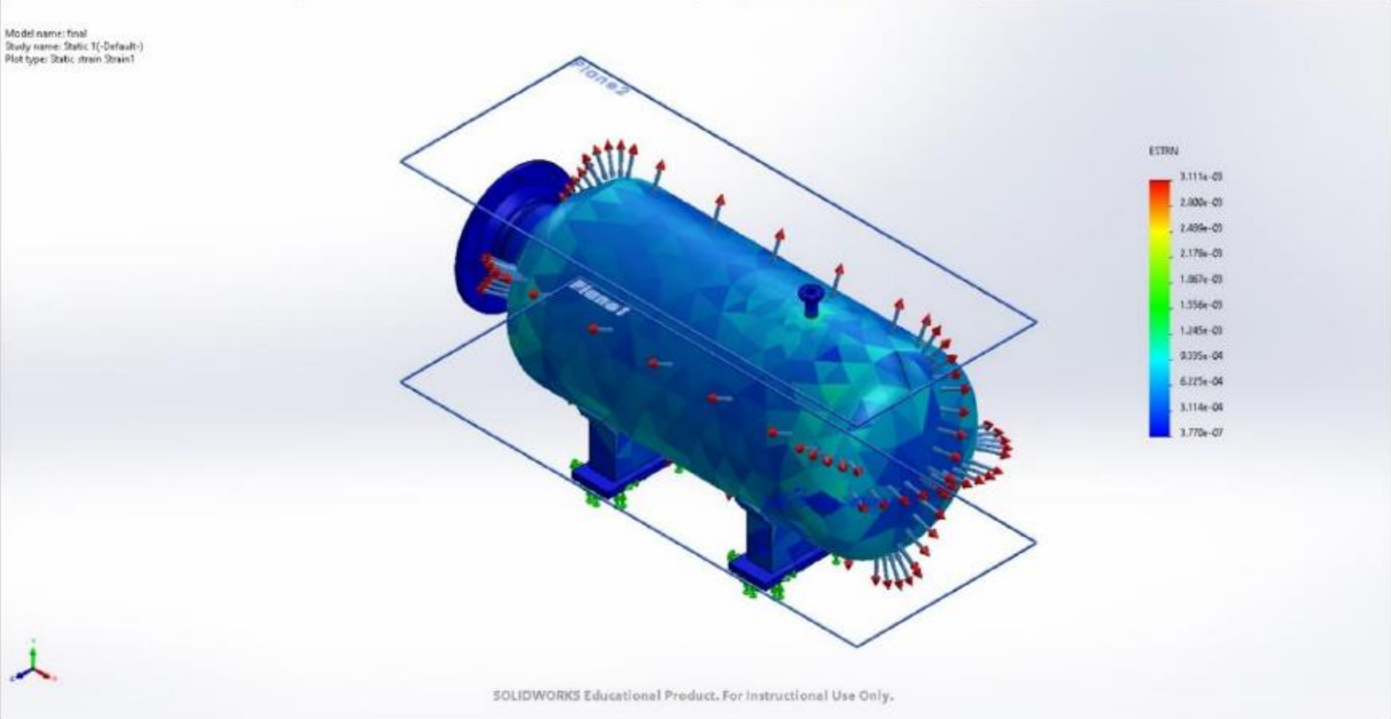


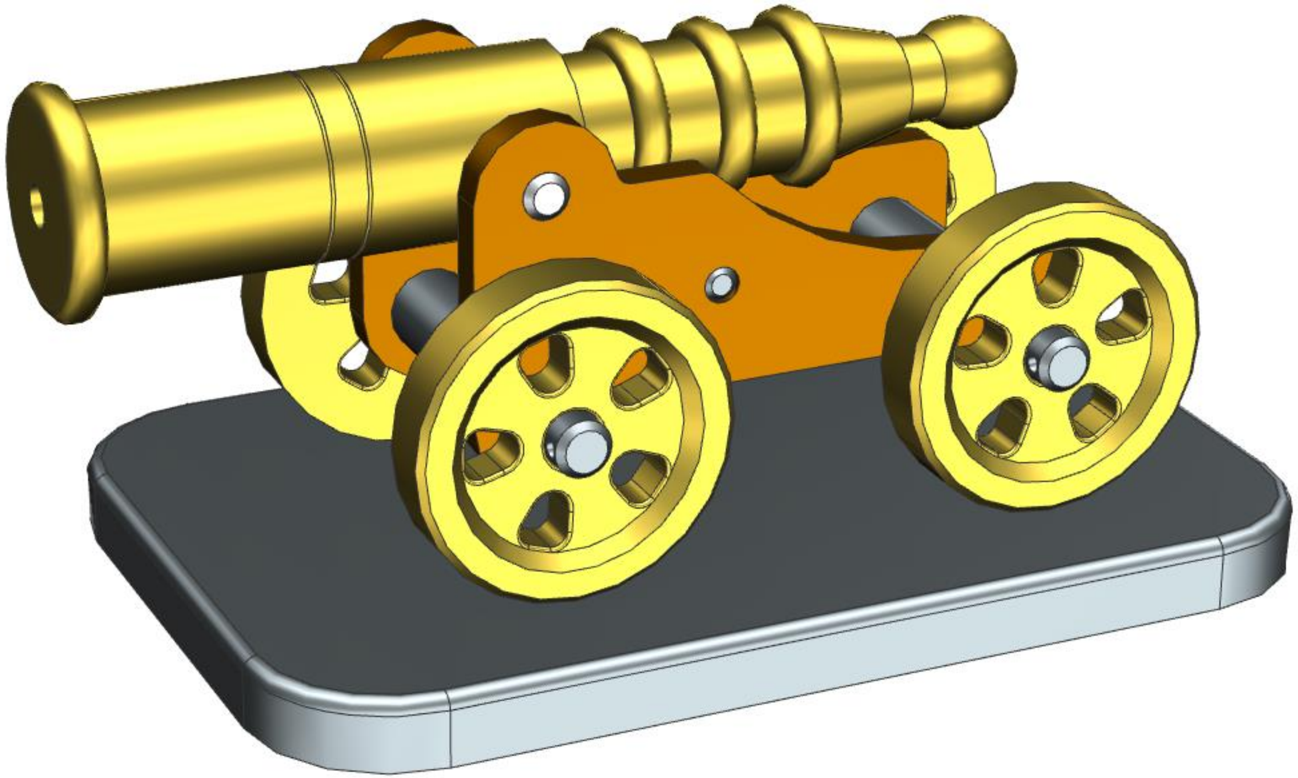
Figure 6: Static 1-Strain-Strain1

- **Key Outcomes:**
  - Verified the vessel's safety through detailed FEA, identifying critical areas for reinforcement. Achieved a robust design capable of withstanding operational pressures with a safety margin. Developed skills in SolidWorks and advanced engineering simulations, contributing to expertise in pressure vessel design.

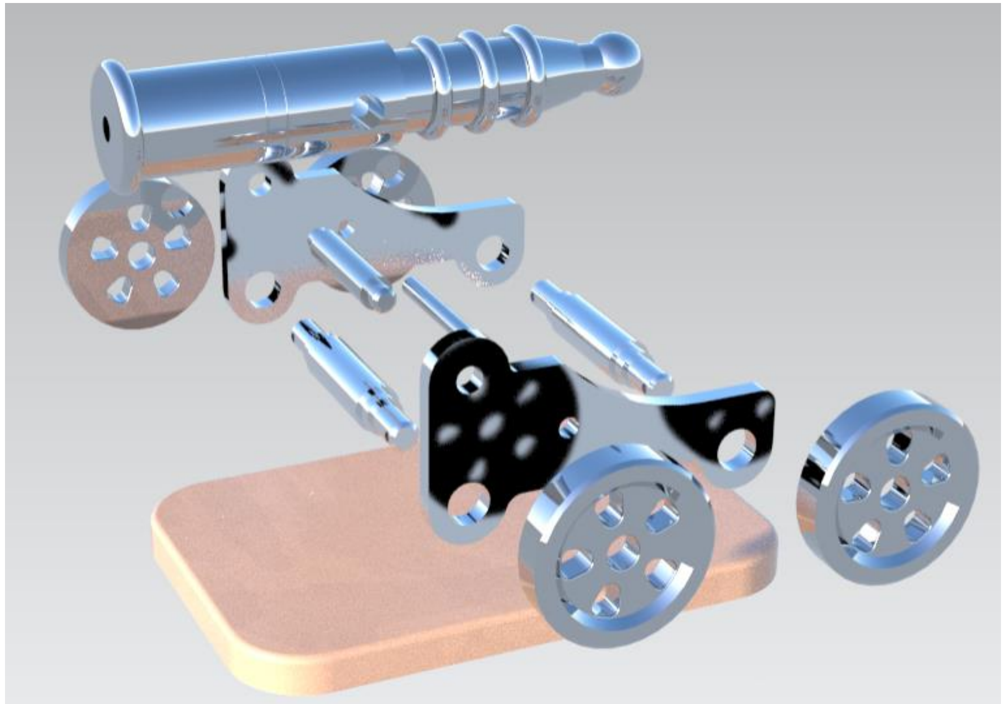
Project Title: Design and Assembly of Wheel Canon

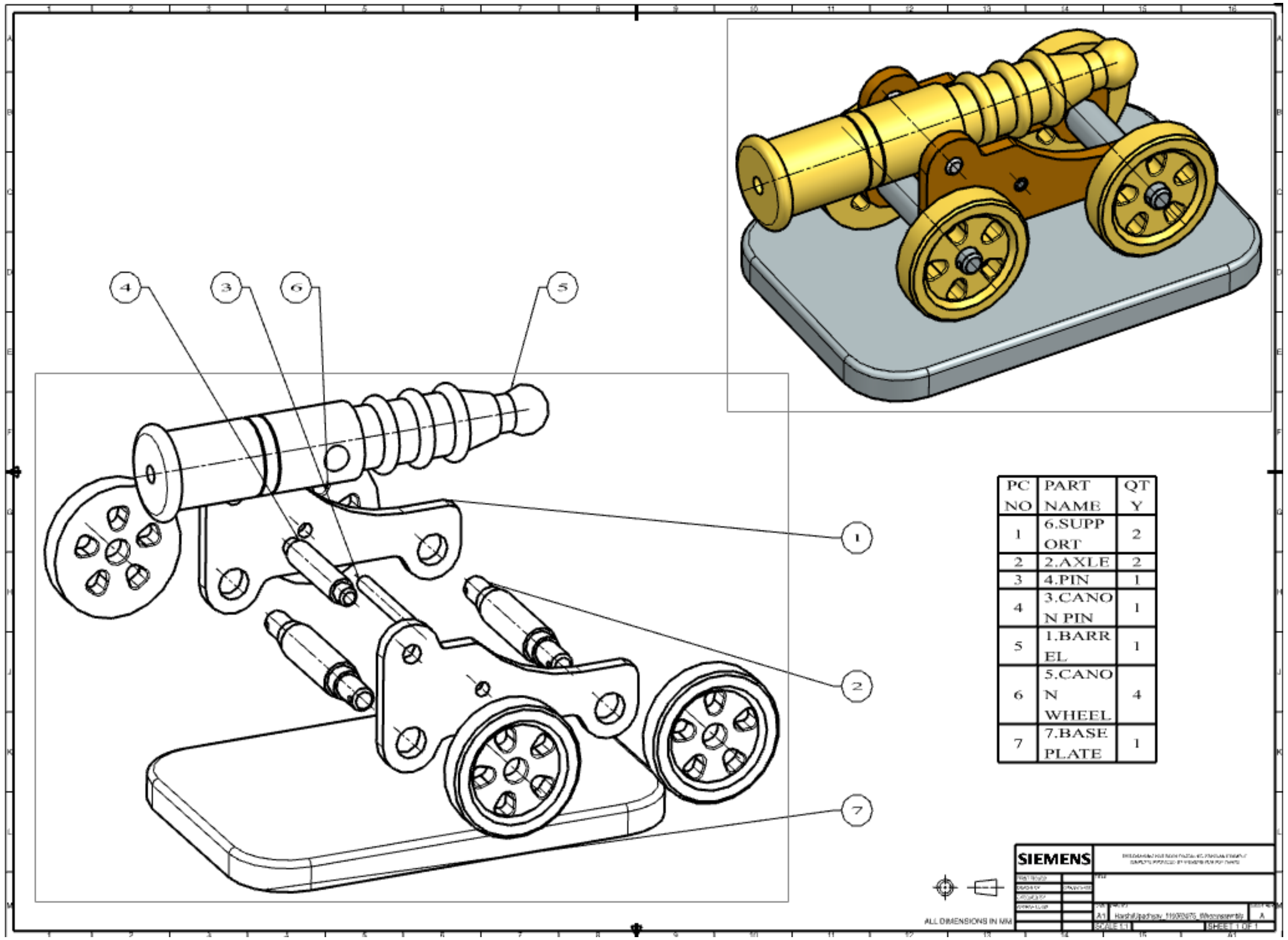
Aim: To design the wheel canon with all its components with assembled final product and with including Drawings.

Approach:



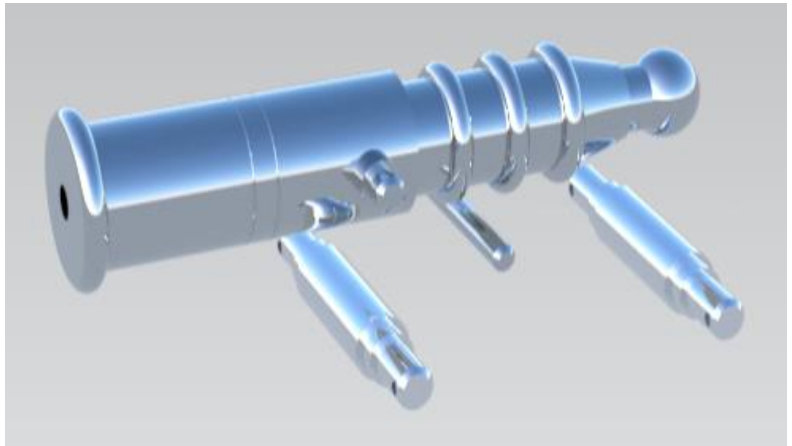
This image displays the render version of exploded view and the drafting of Exploded view with BOM (Bill Of Material).

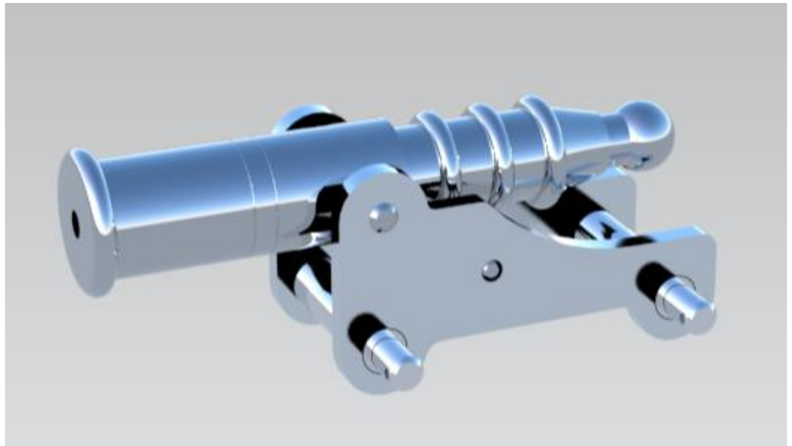


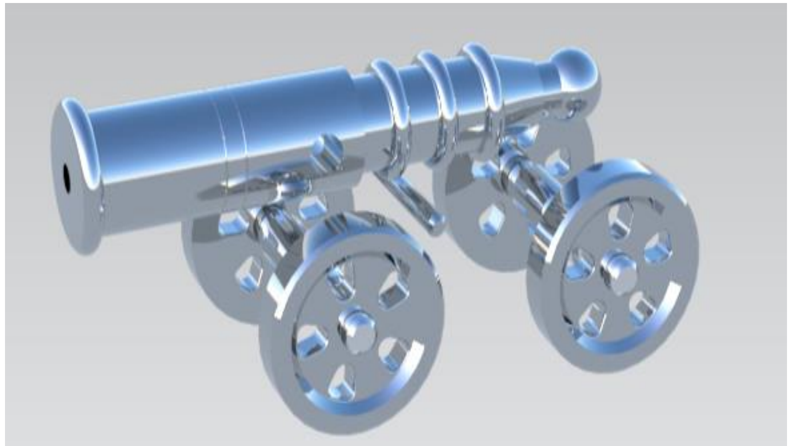


PC NO	PART NAME	QTY
1	6.SUPPORT	2
2	2.AXLE	2
3	4.PIN	1
4	3.CANON PIN	1
5	1.BARREL	1
6	5.CANON WHEEL	4
7	7.BASE PLATE	1

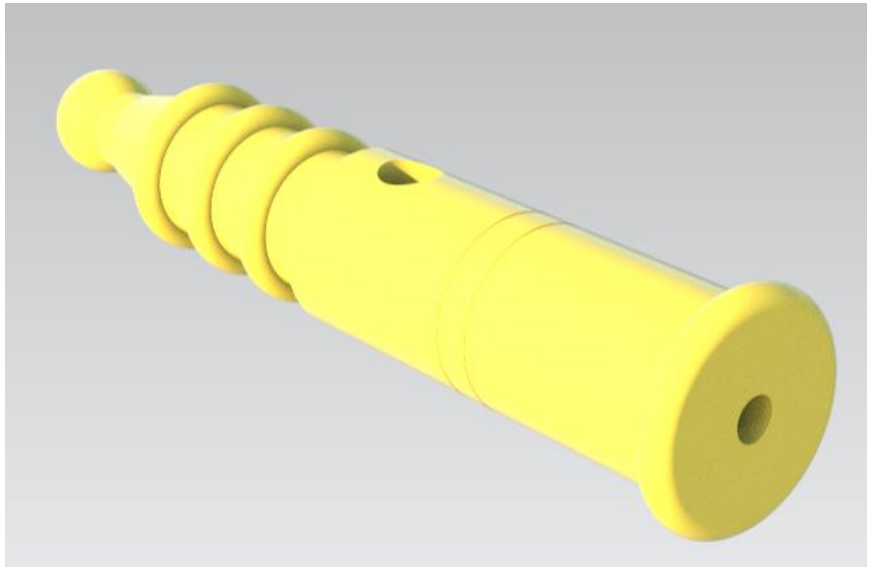
Some of the render version image without support, canon wheel ,and support and wheel.

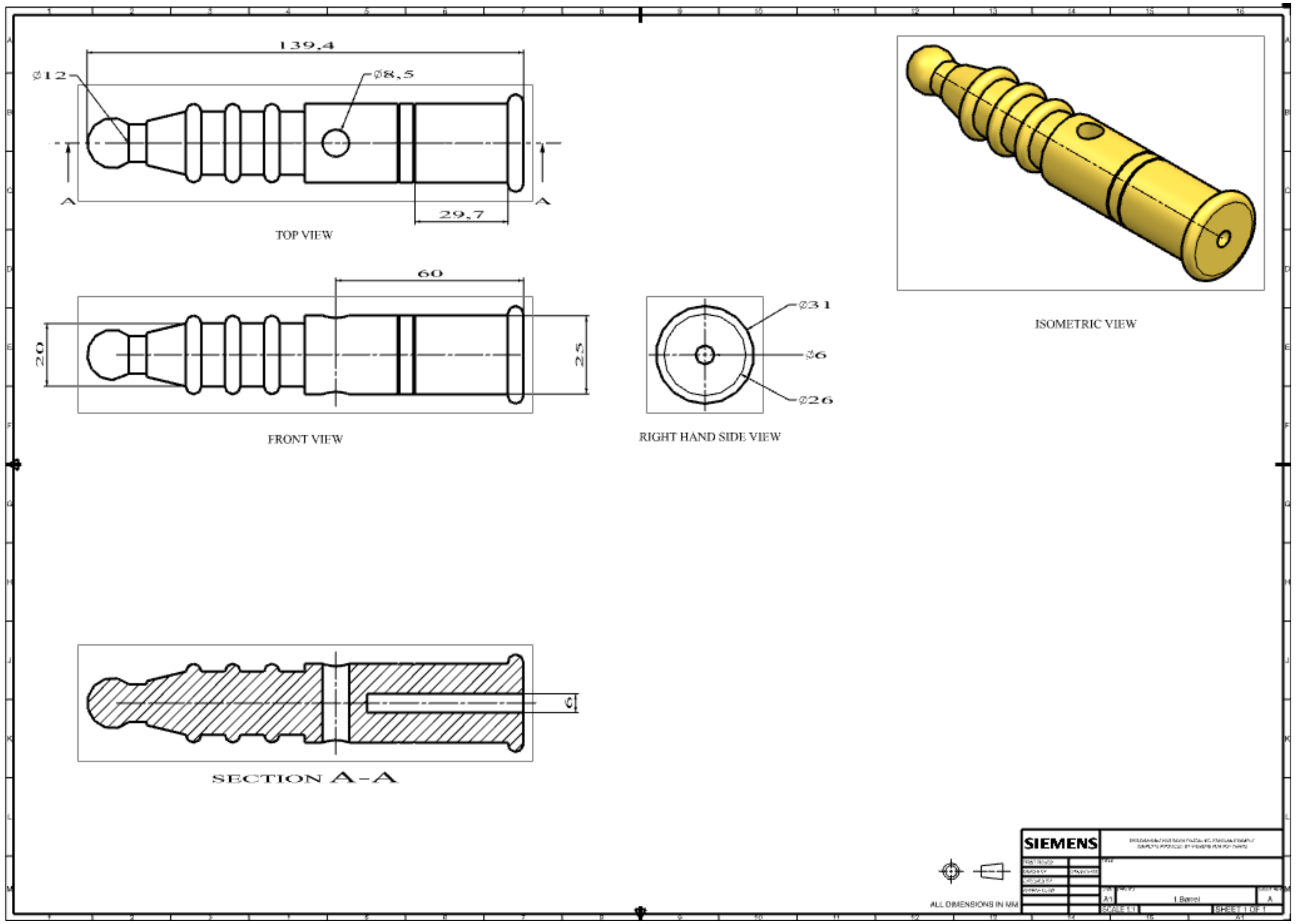




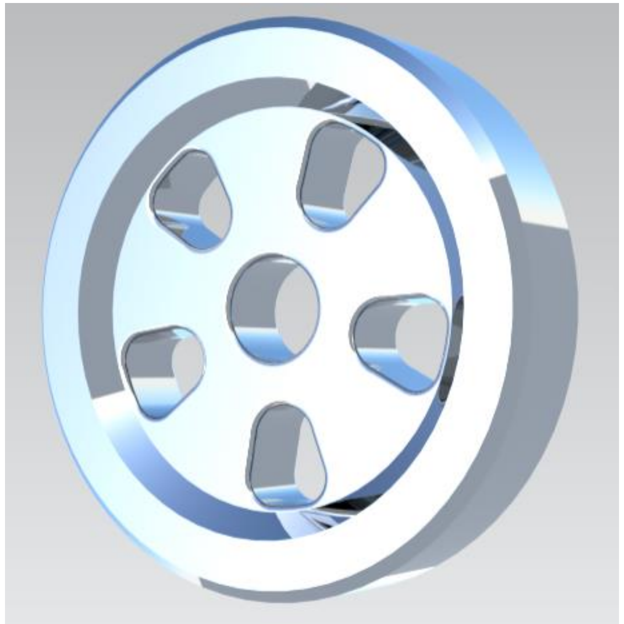


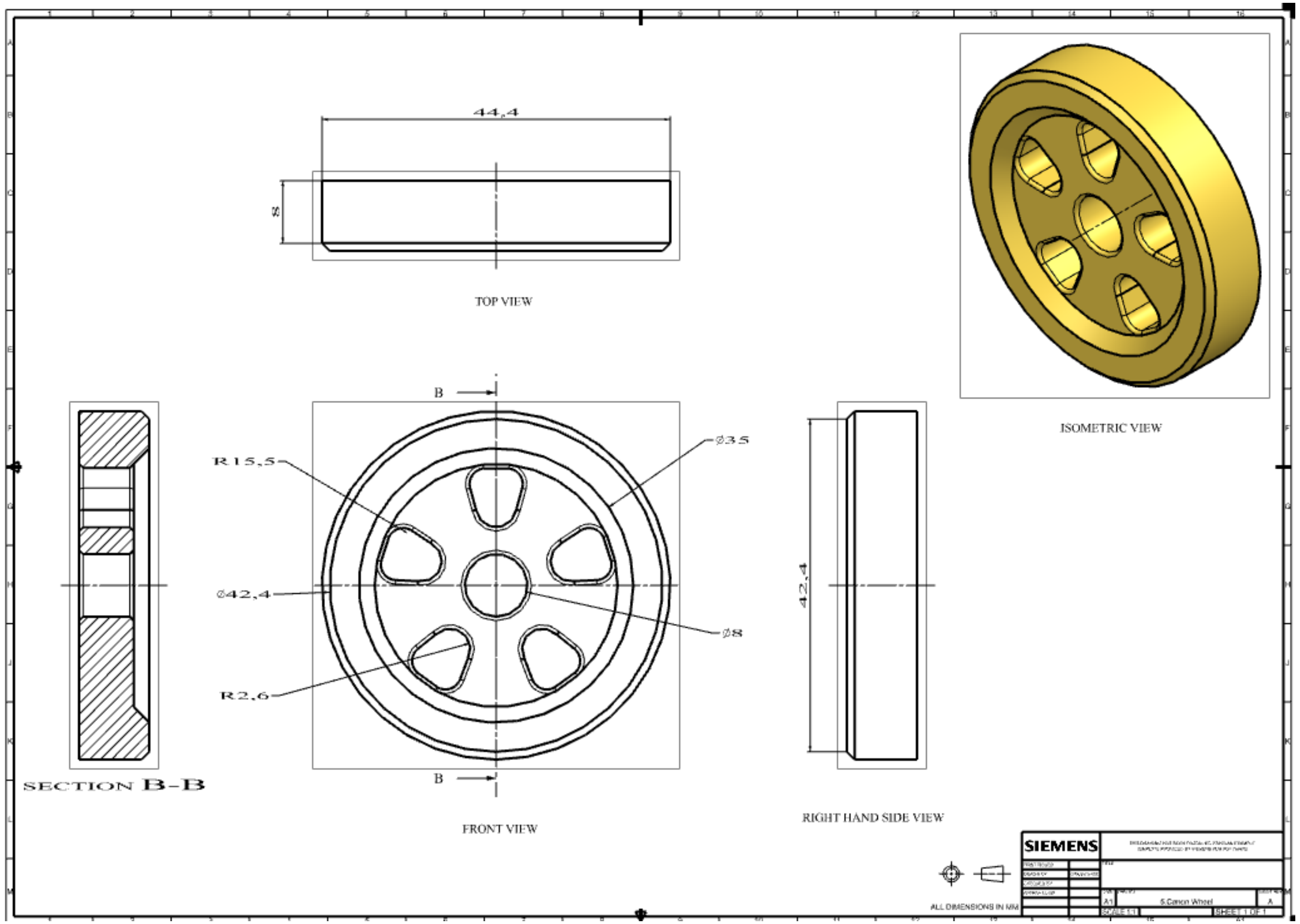
Render version image of the barrel and drafting sheet with the 3rd angle projection view system and section view.





Canon wheel drafting sheet with section view and render version of image.

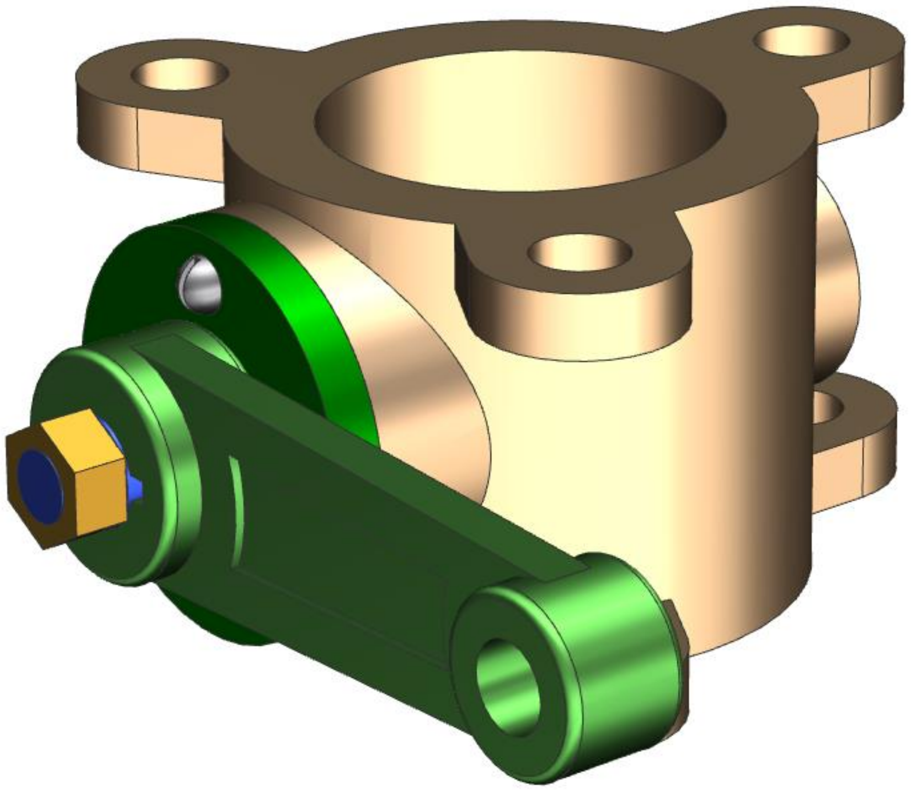




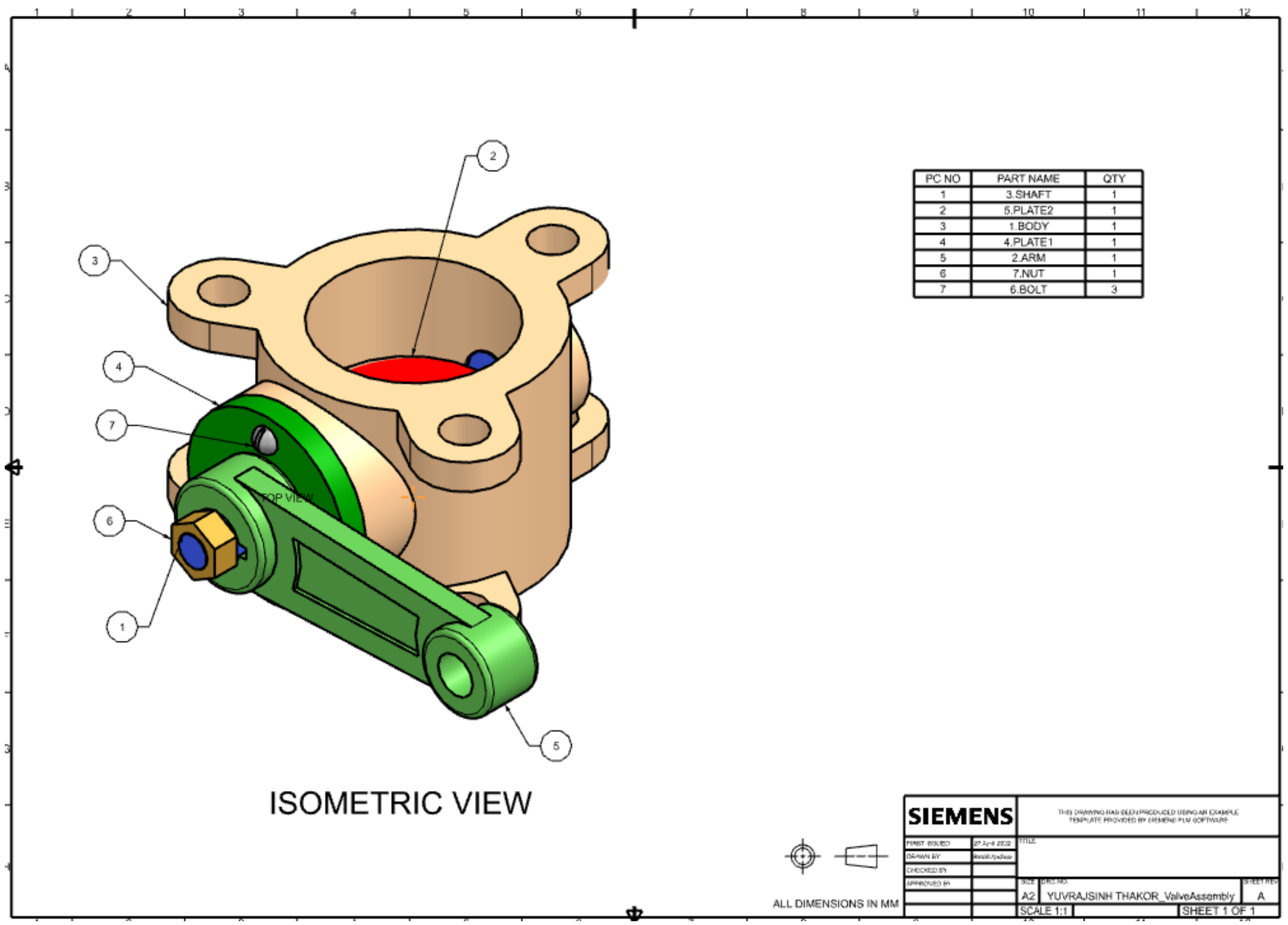
Project Title: Design and Assembly of Valve

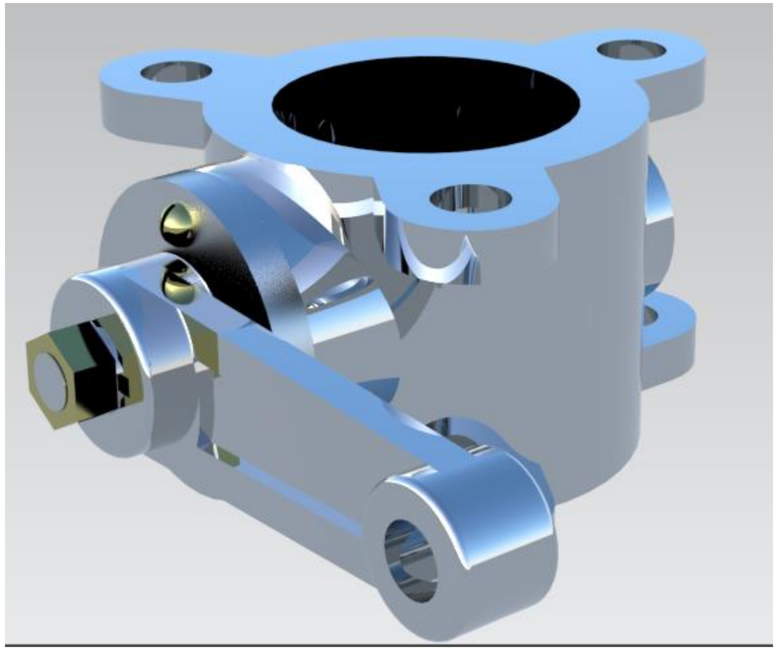
Aim: To design the valve with all its components with assembled final product and with including Drawings.

Approach:

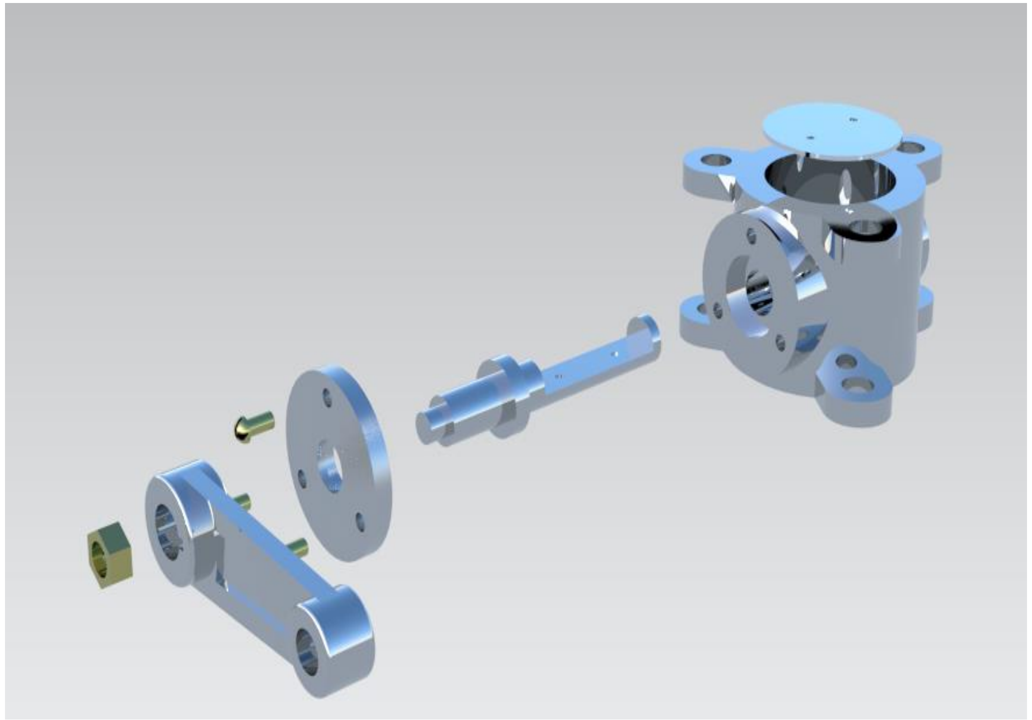


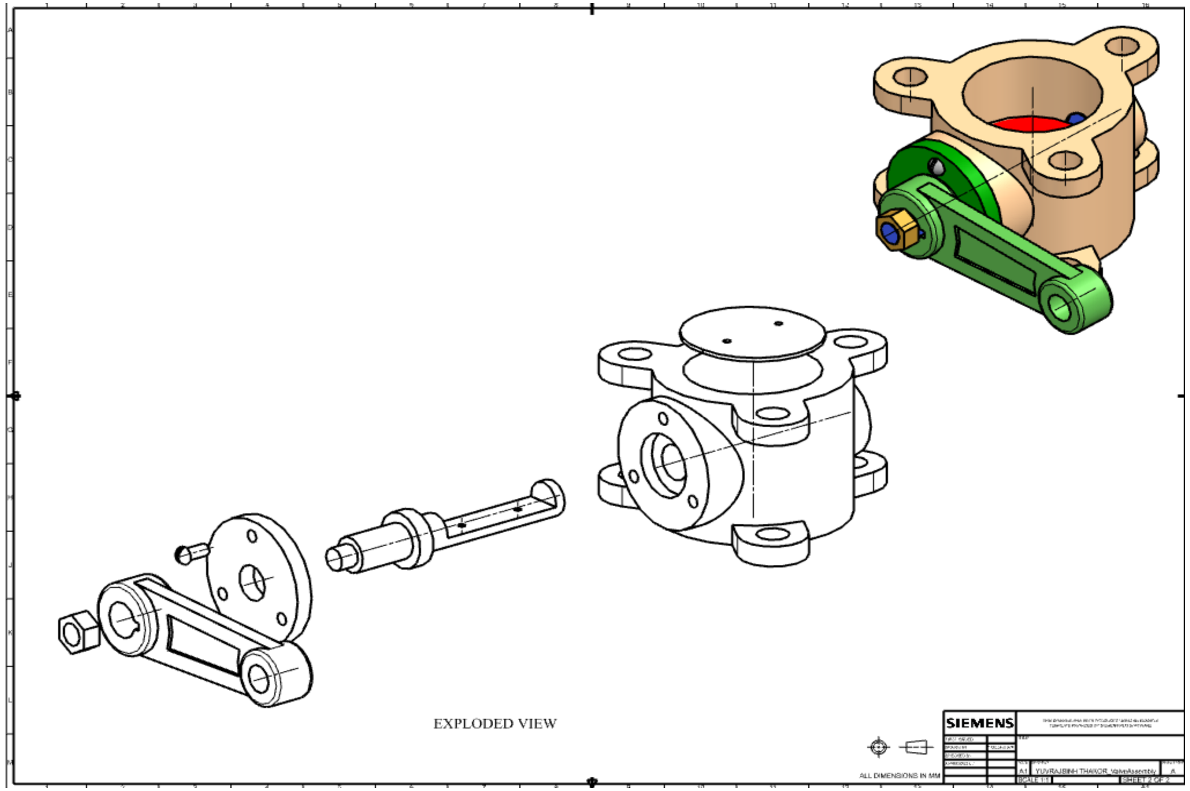
Drafting function with BOM (Bill Of Material) and render image of main assembly.

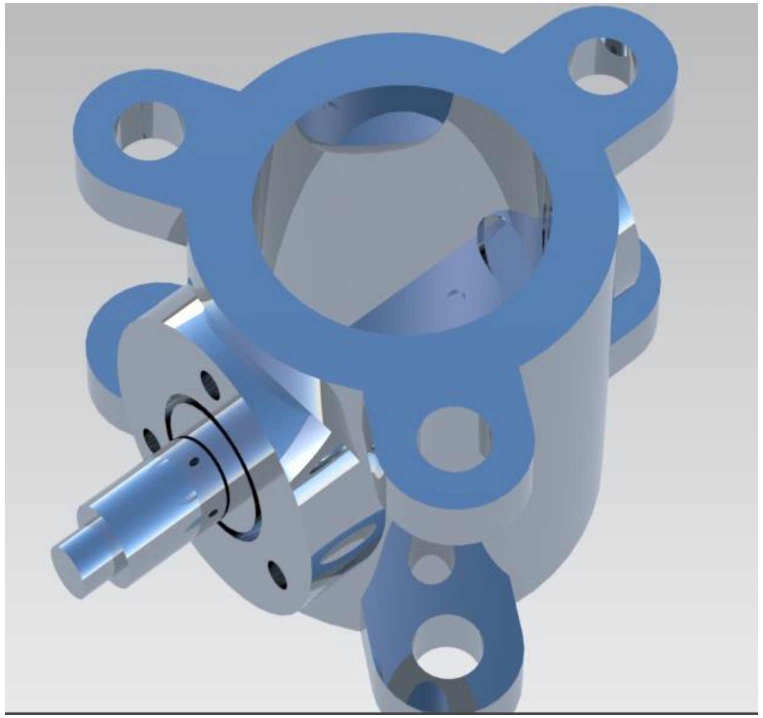




Render version image of the Exploded view with the Drafting sheet.

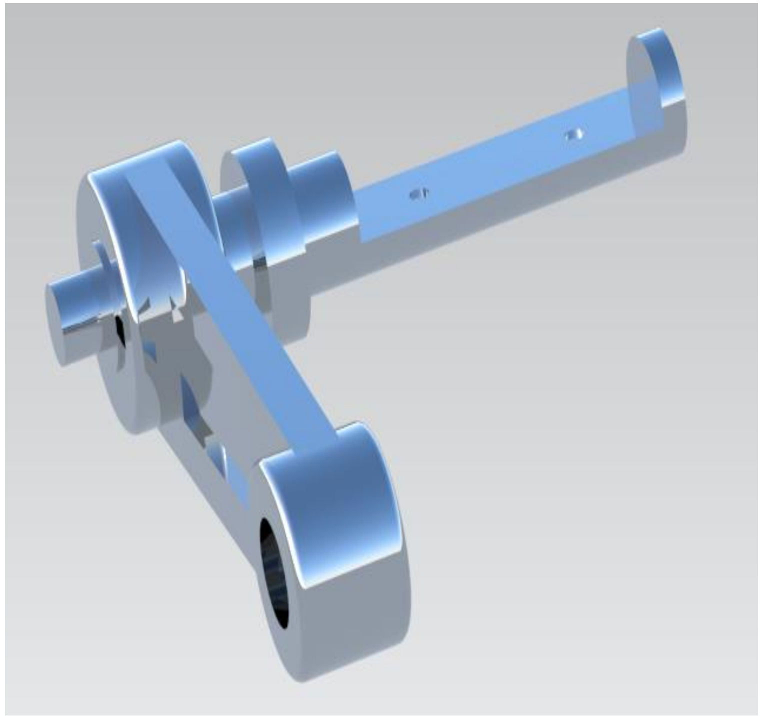


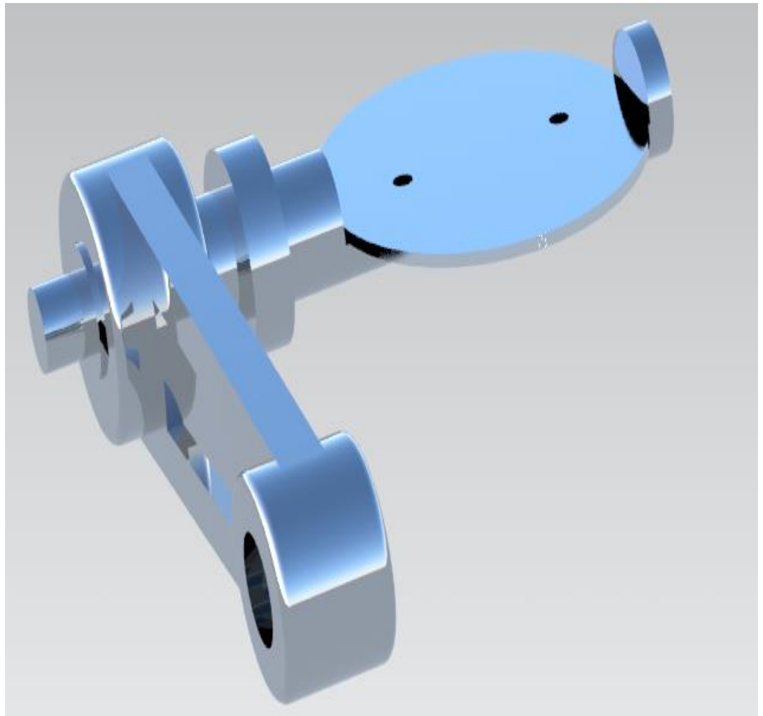




Render version image without arm and plate.

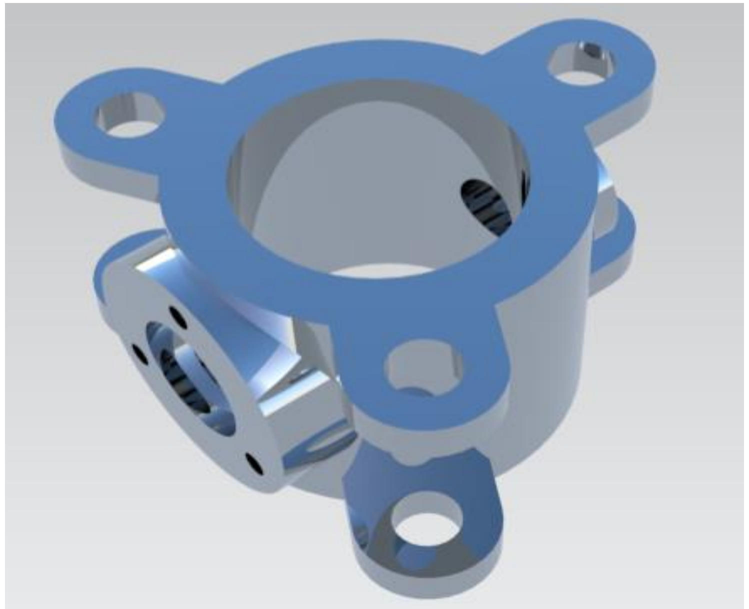
Render version image without body, plate, nut and bolt.

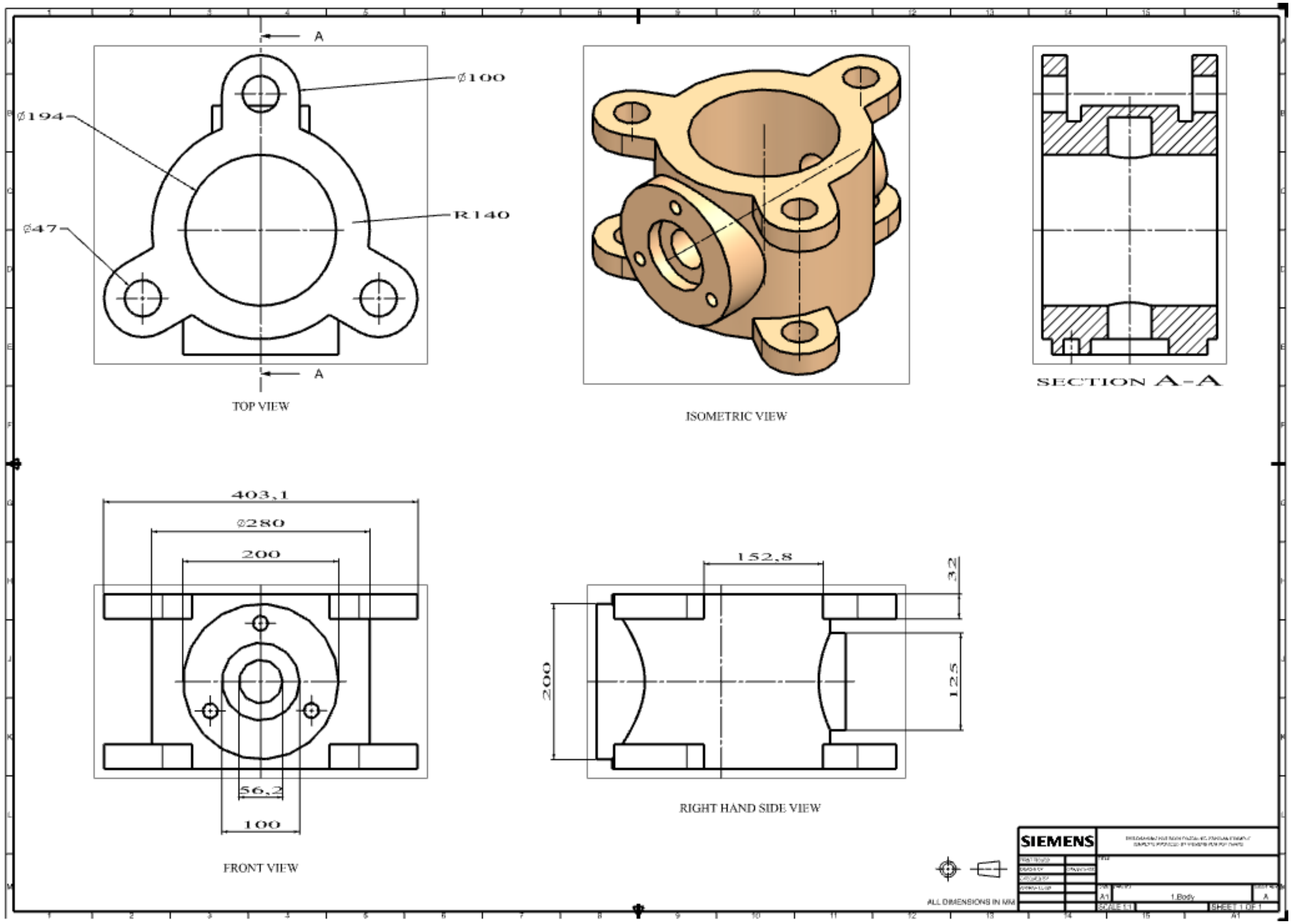




Render version image without body, nut and bolt.

Main body with render version image and drafting sheet having section and all three views.





Drafting function for Arm with different view and section view.